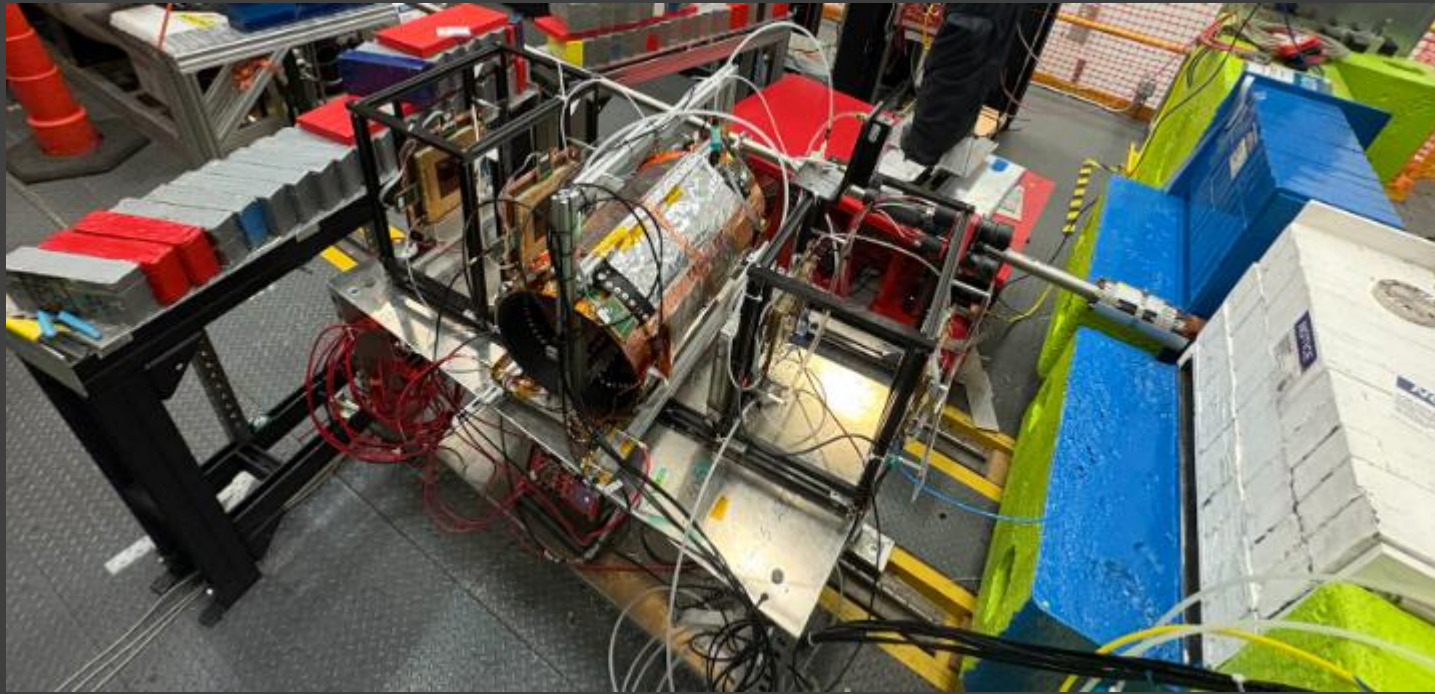




Mapping for a Cylindrical Micro-Resistive Well Detector

Milestone 4

People

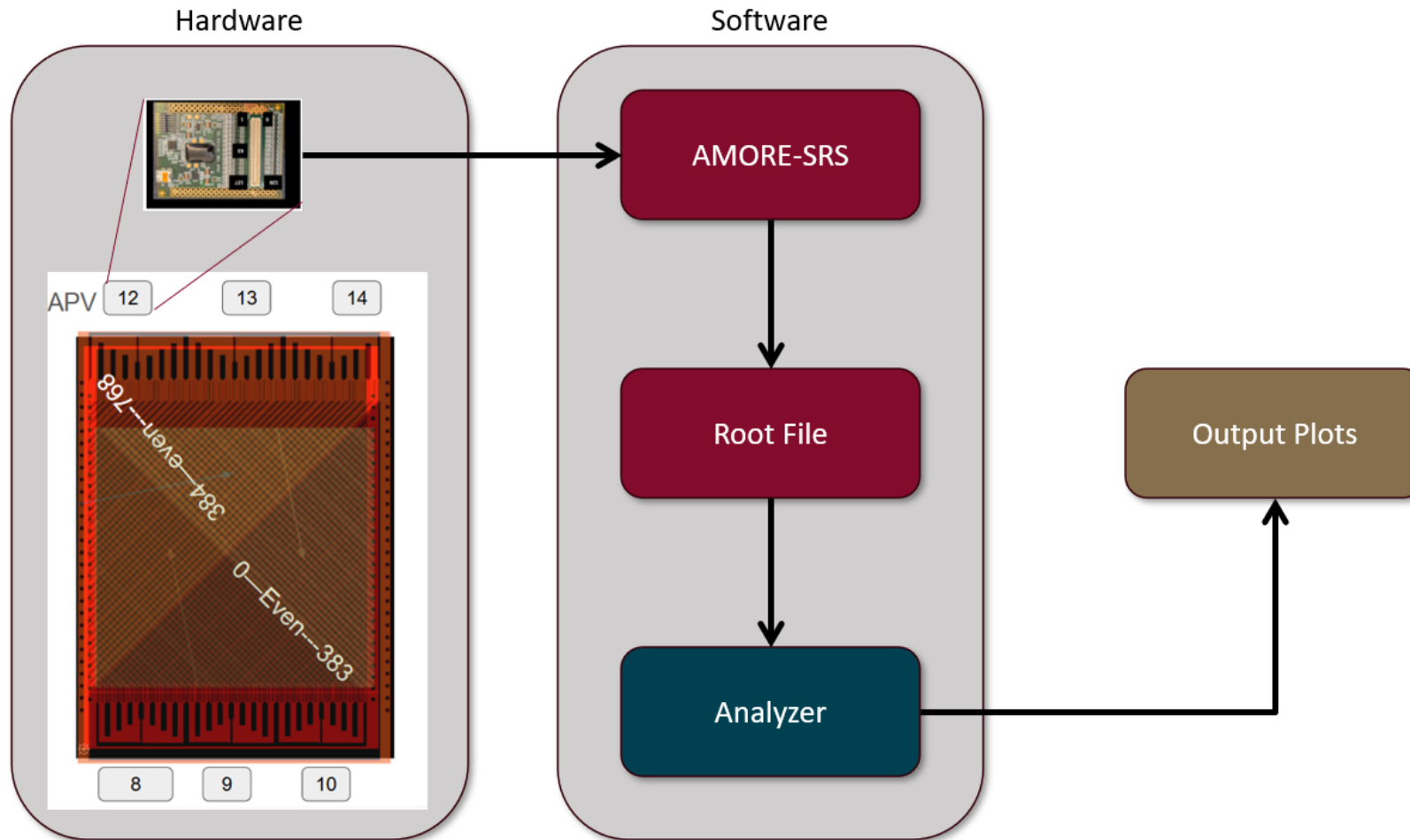
Group Members

- Alejandro Moreno Zuluaga,
moreno2023@my.fit.edu
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smueller2023@my.fit.edu

Faculty Advisors/Clients

- Pietro Iapozzuto,
piapozzuto2015@my.fit.edu
- Dr. Marcus Hohlmann,
hohlmann@fit.edu

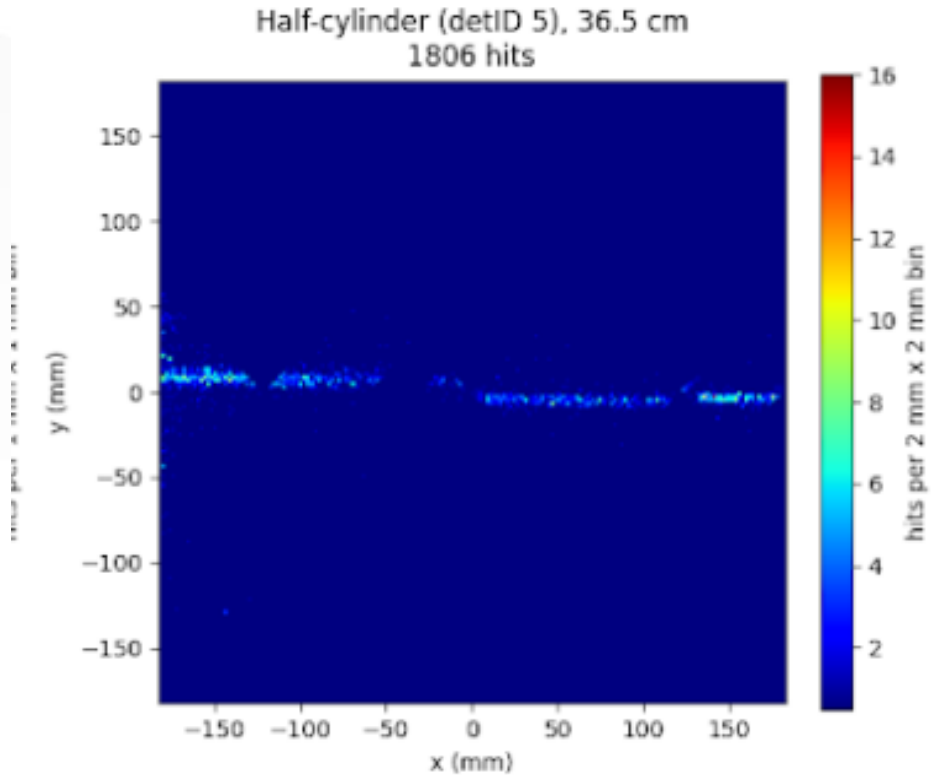
Overview



Milestone 4 Task Matrix

Task	Completion	Alejandro	Summer	TO DO
I. Find correct cylindrical mapping	90%	90%	0%	Current reconstruction shows clear beam but can't confirm until compare with trackers
II. Create efficiency plots	80%	0%	80%	Need to apply clustering and iterative process for shifting the offset
III. Compare cylindrical detectors and trackers	20%	10%	10%	Need to compare detector data to tracker data using the calculated trajectories

Task- Cylindrical Mapping

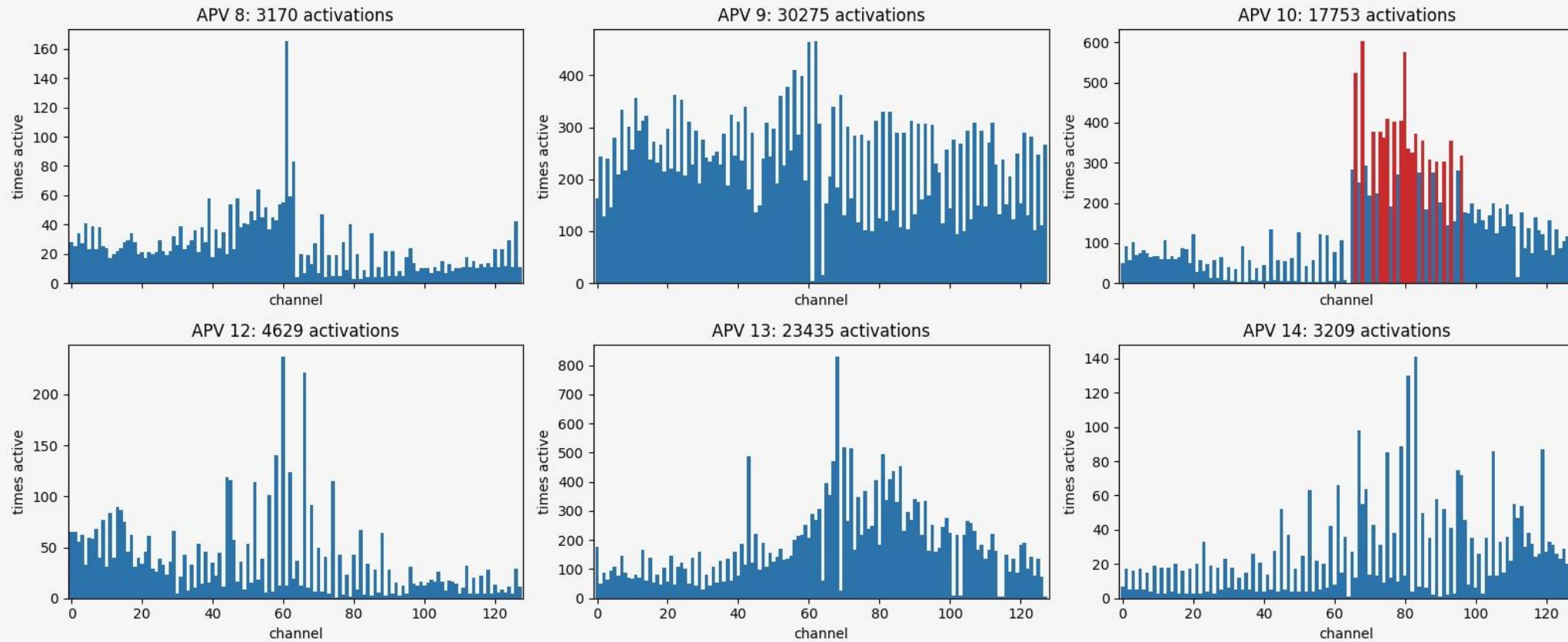


Mapping function:
 $(\text{Channel} + 64) \% 128$

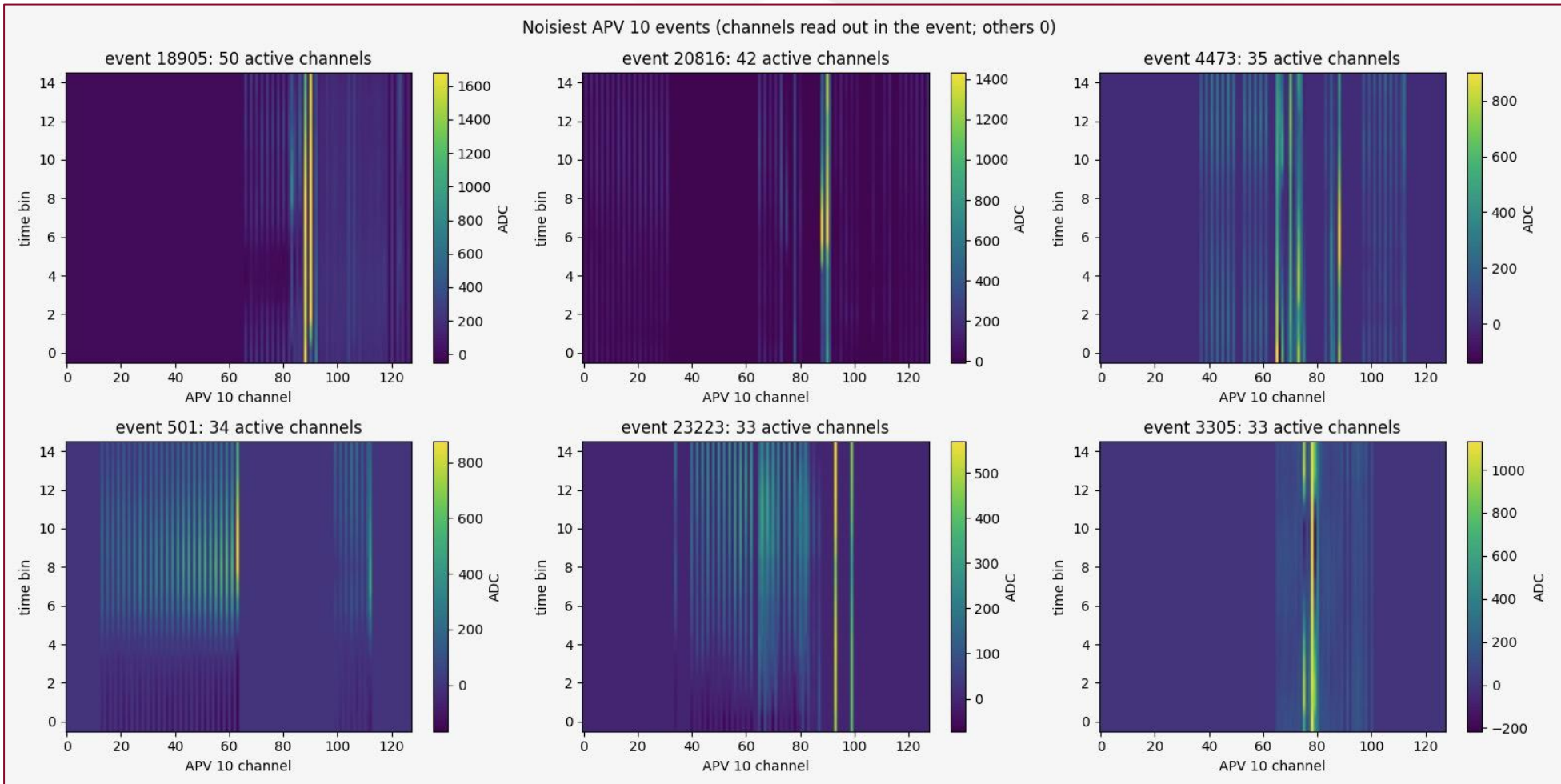
UV split after mapping:
Us = even
Vs = odd

Task- Cylindrical Mapping

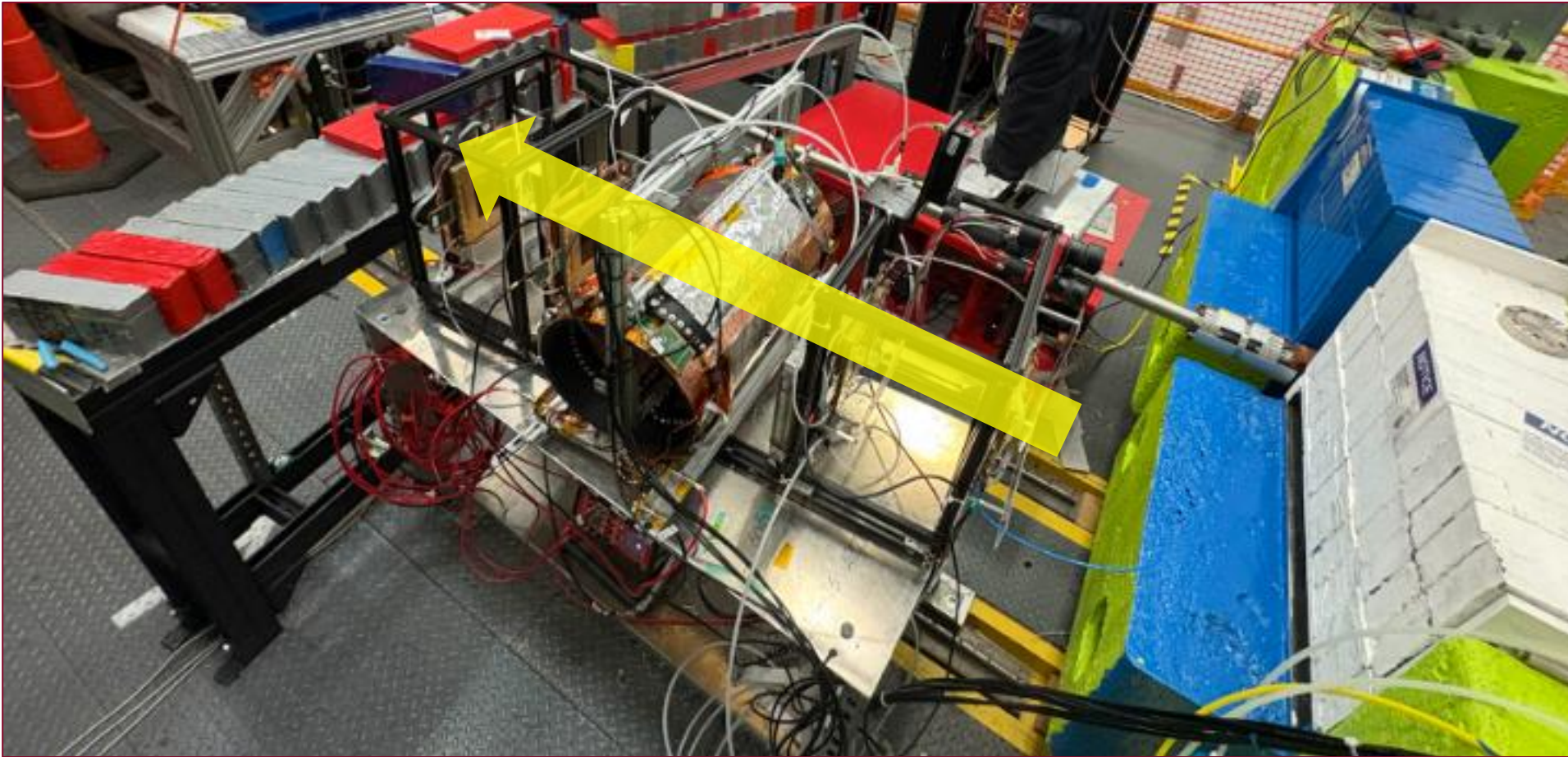
Detector 5: channels passing the ADC filter (peak > 80), all events (red = noisy APV 10 channels)



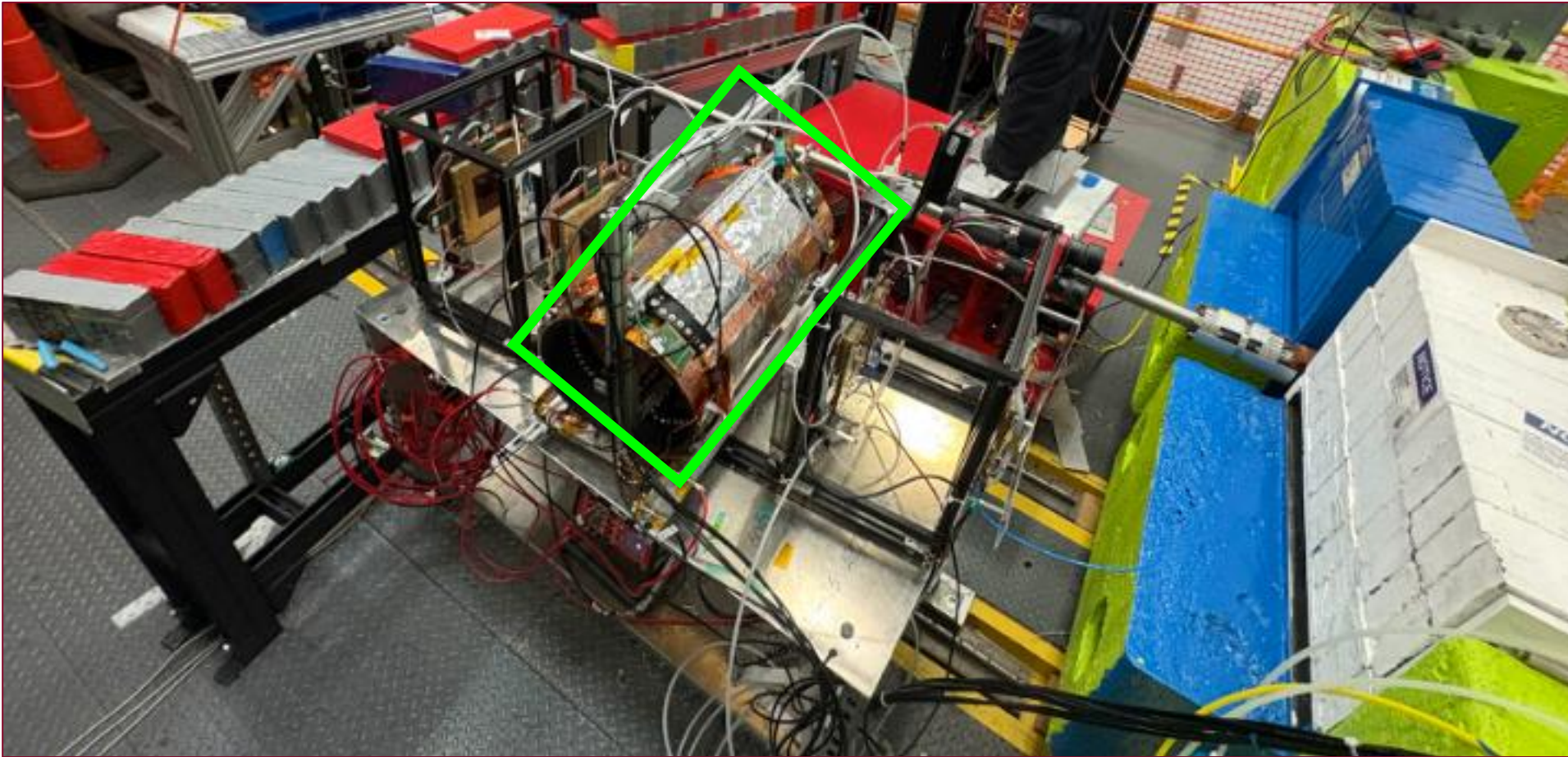
Task- Cylindrical Mapping



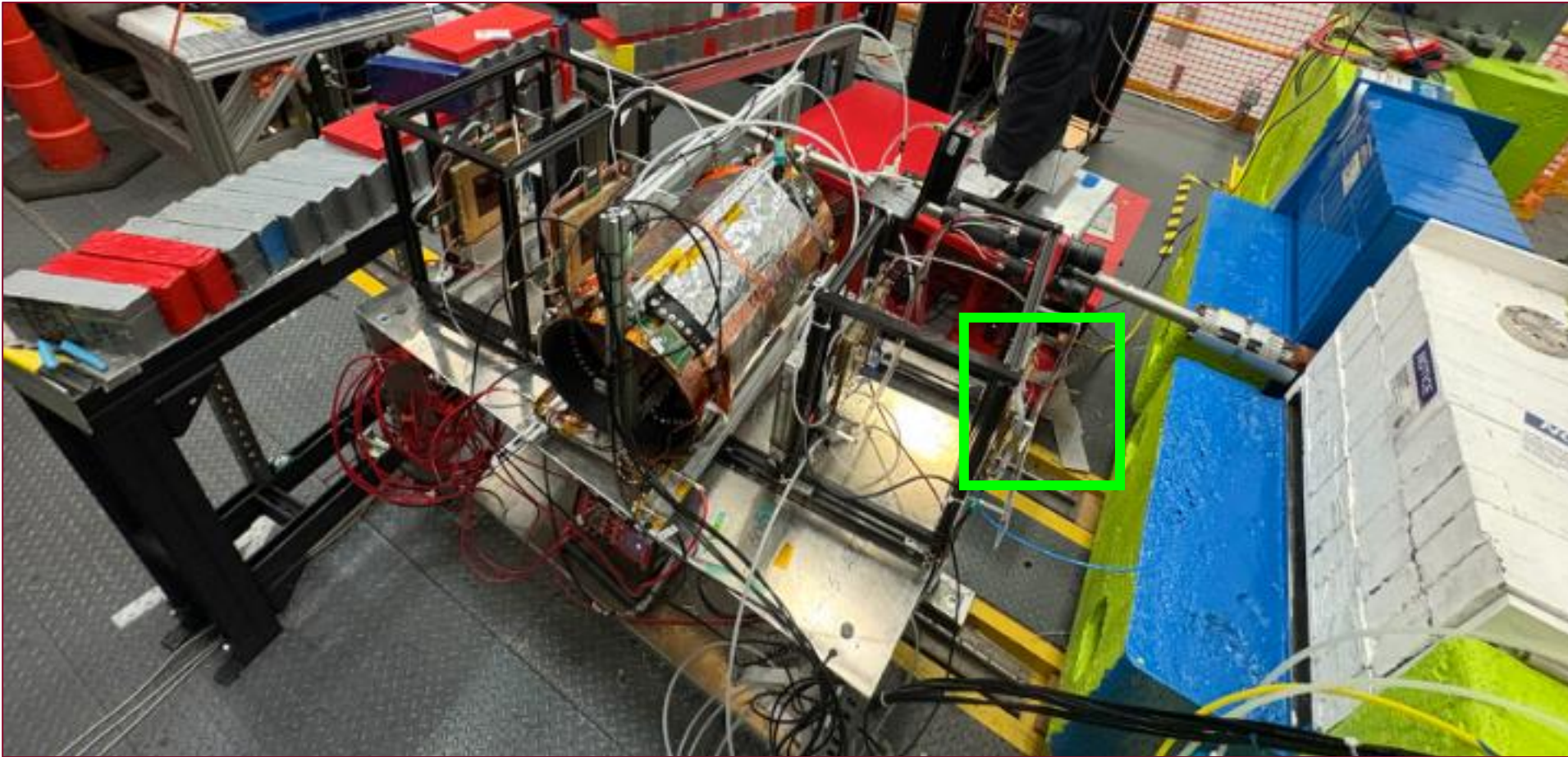
Task- Create Efficiency Plots



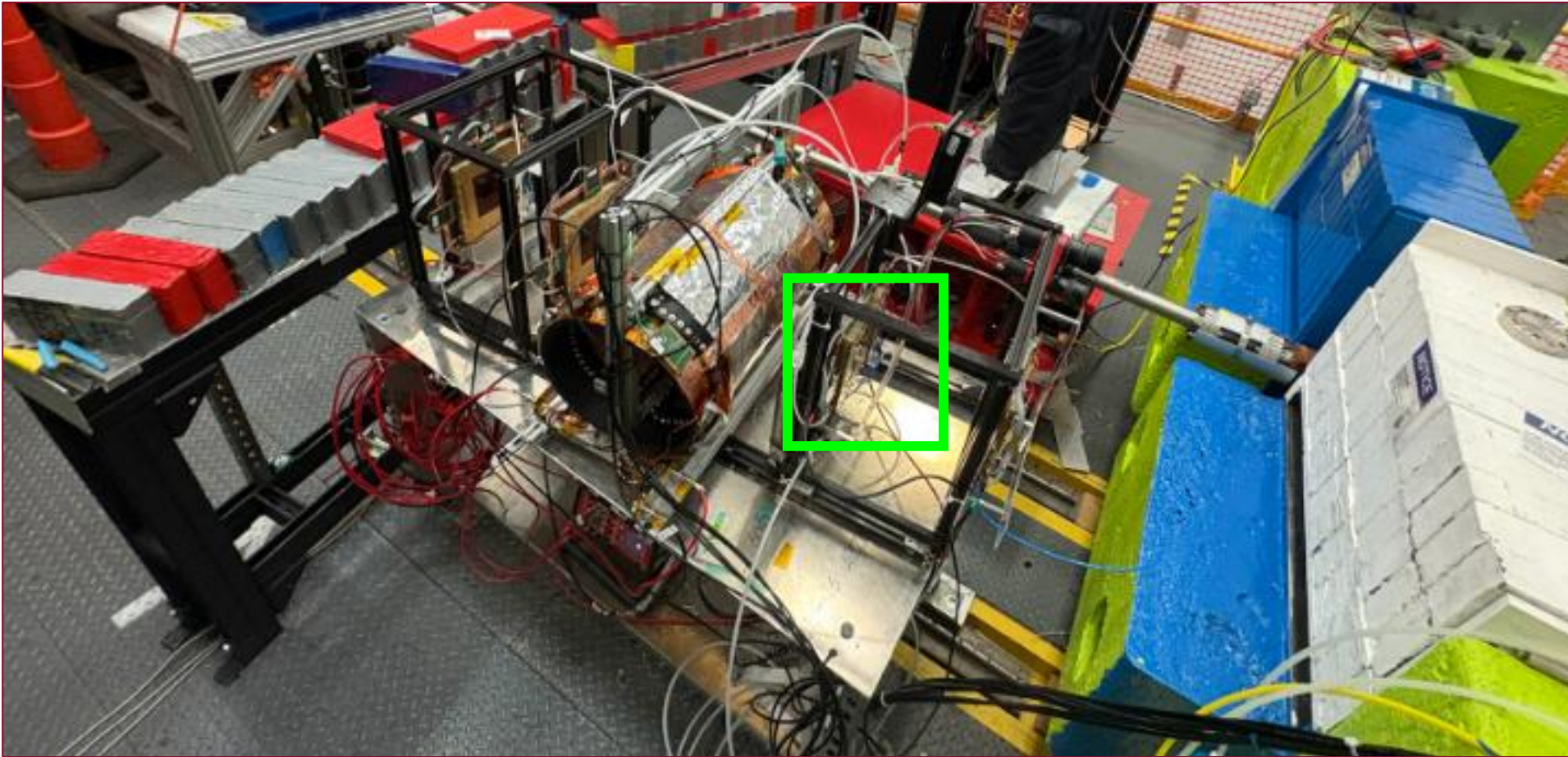
Task- Create Efficiency Plots



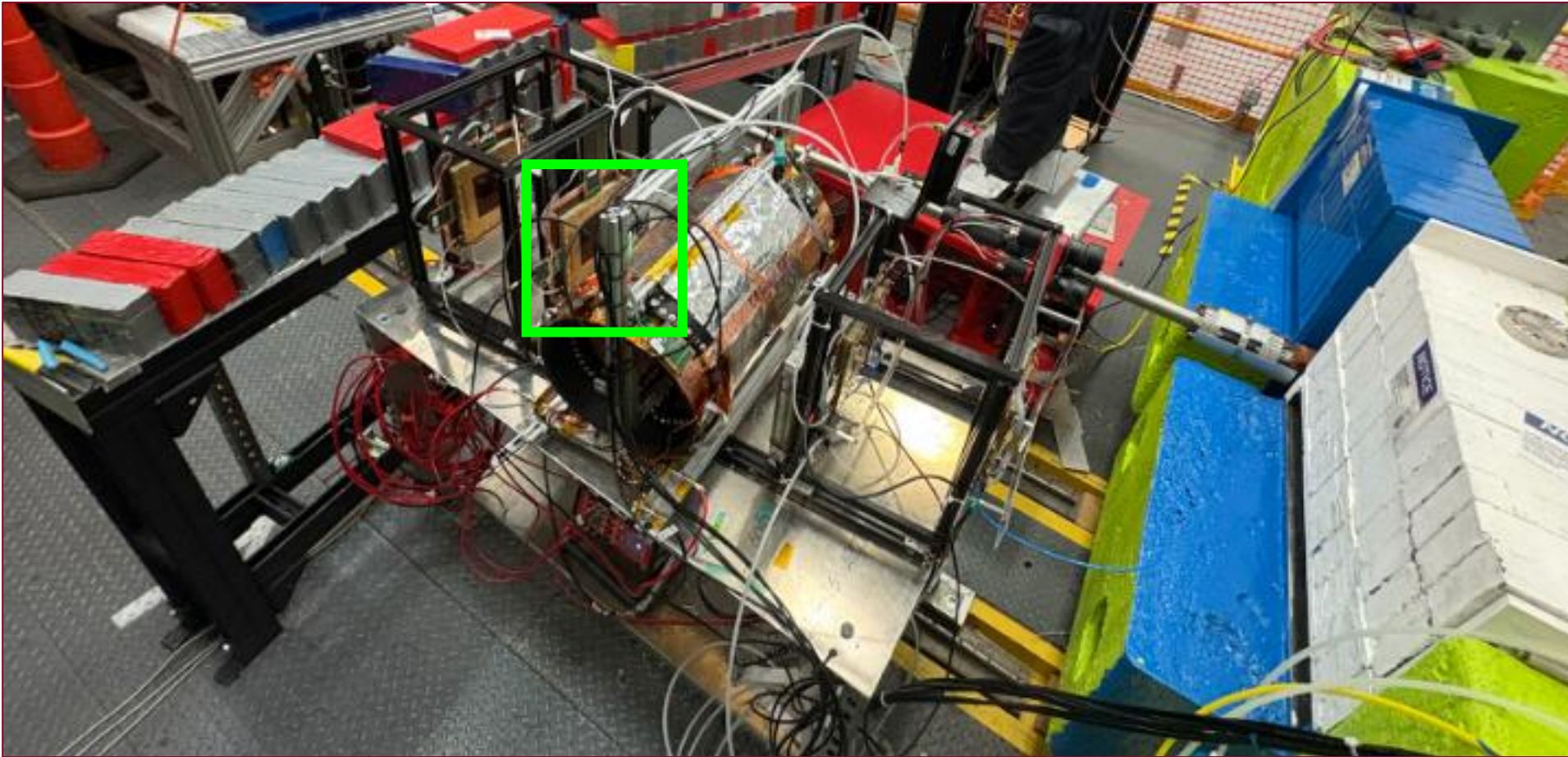
Task- Create Efficiency Plots



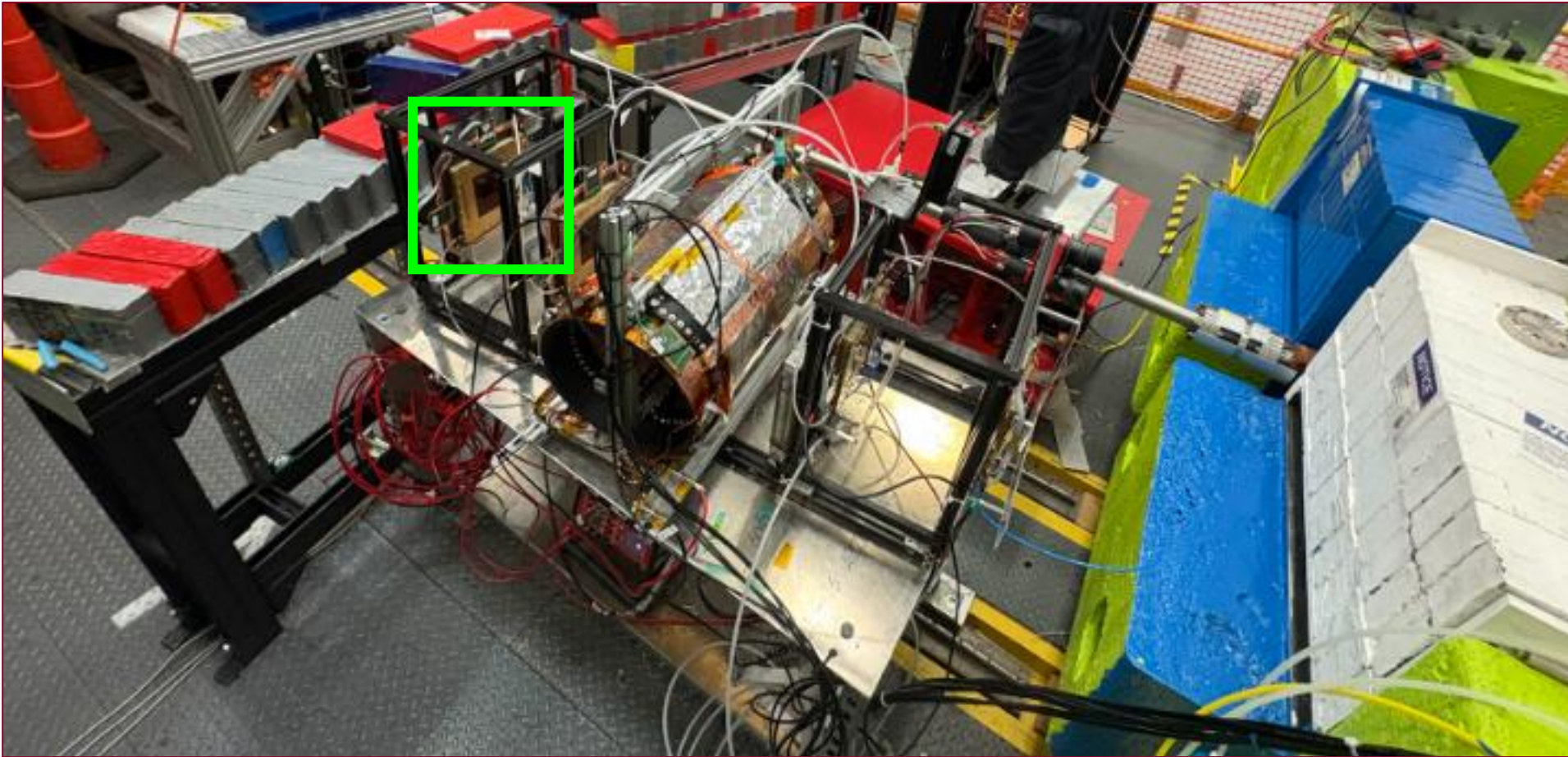
Task- Create Efficiency Plots



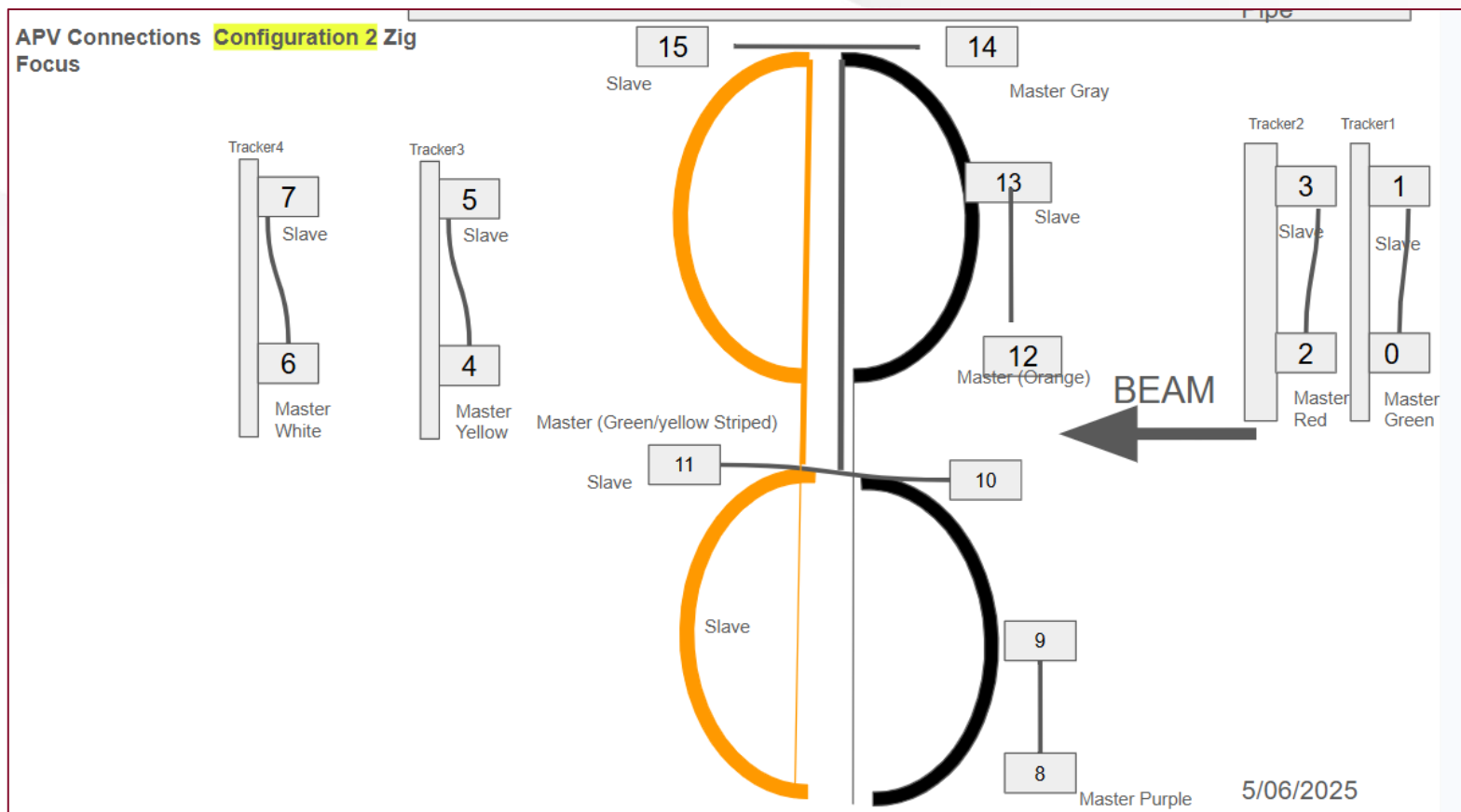
Task- Create Efficiency Plots



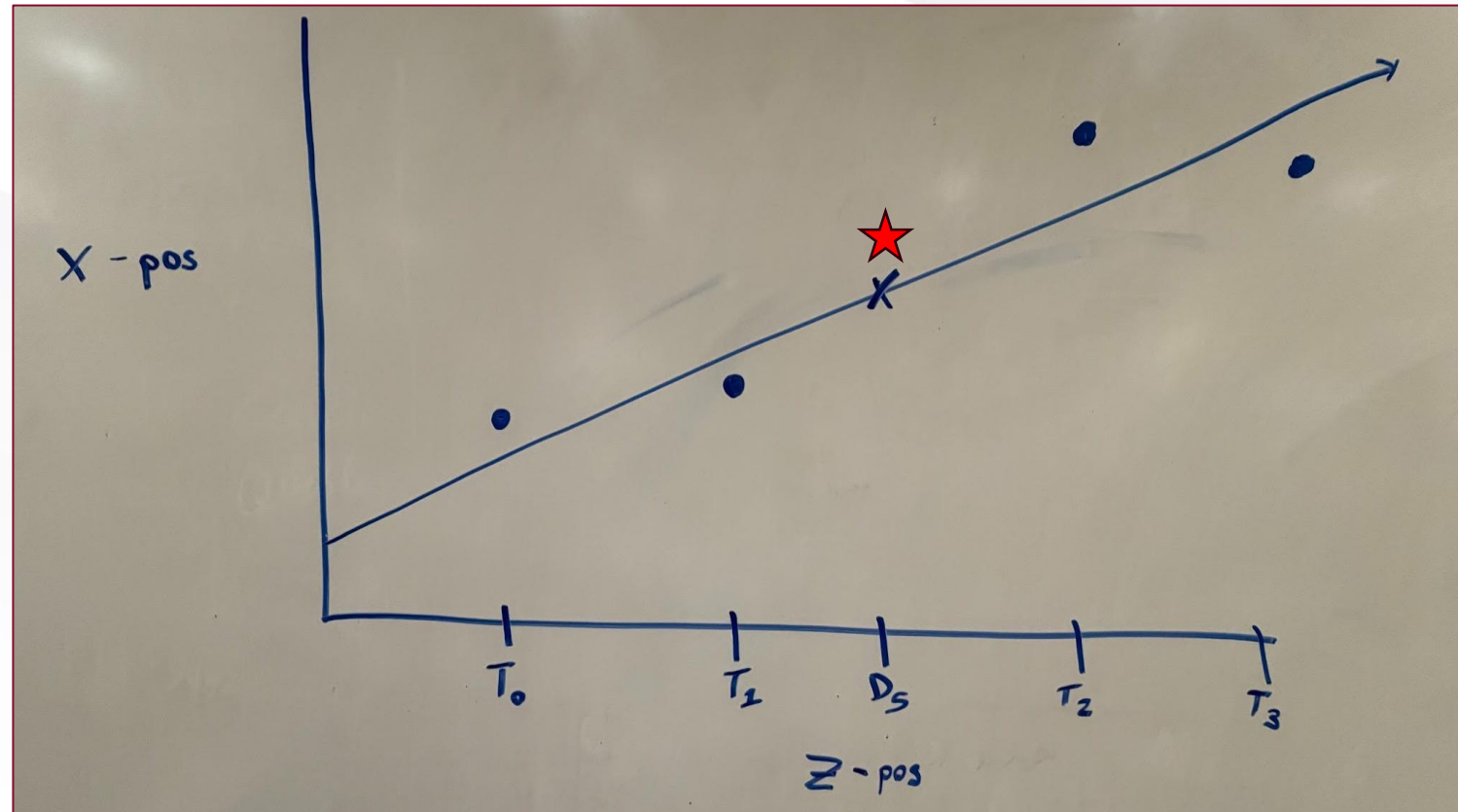
Task- Create Efficiency Plots



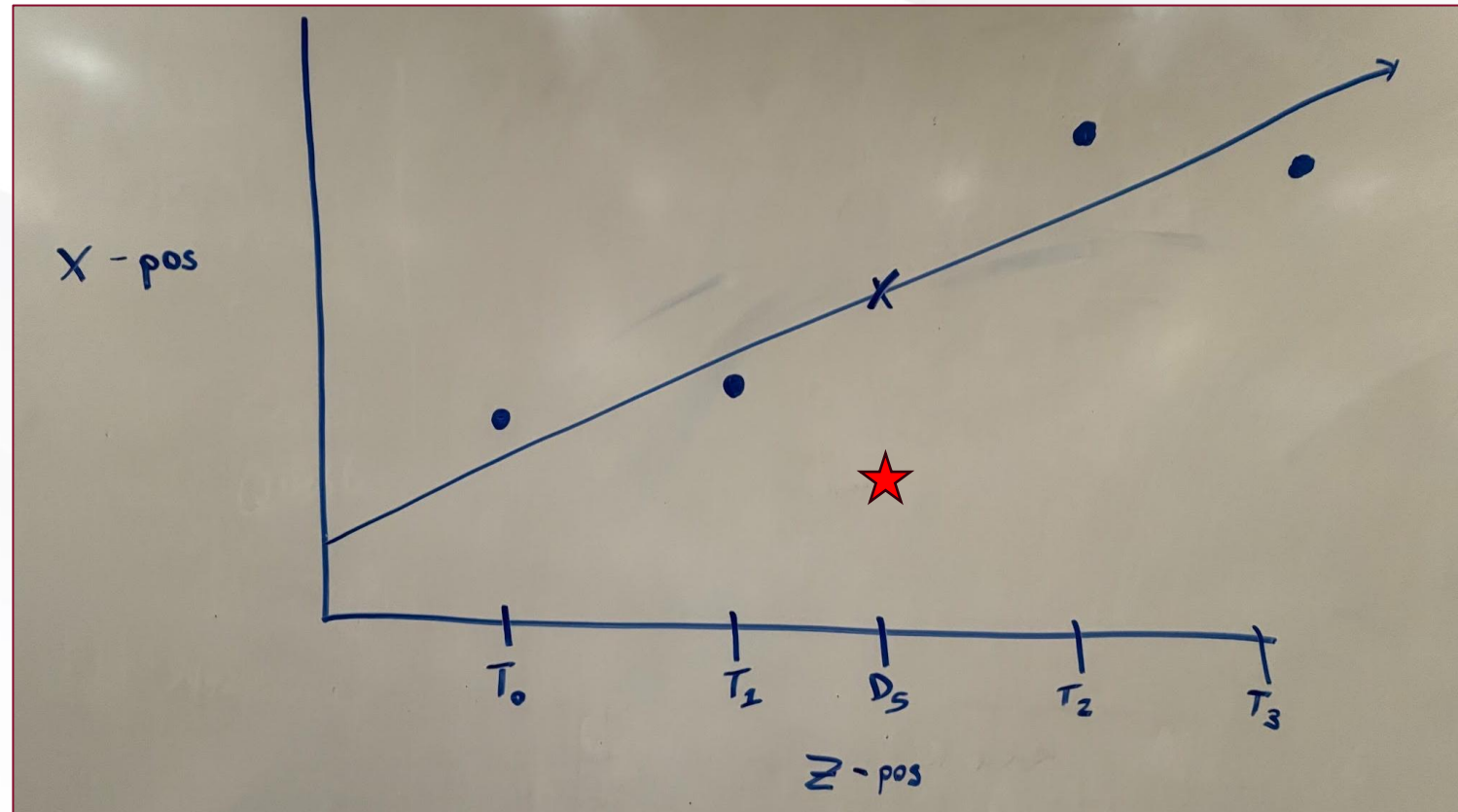
Task- Create Efficiency Plots



Task- Create Efficiency Plots



Task- Create Efficiency Plots



Task- Create Efficiency Plots

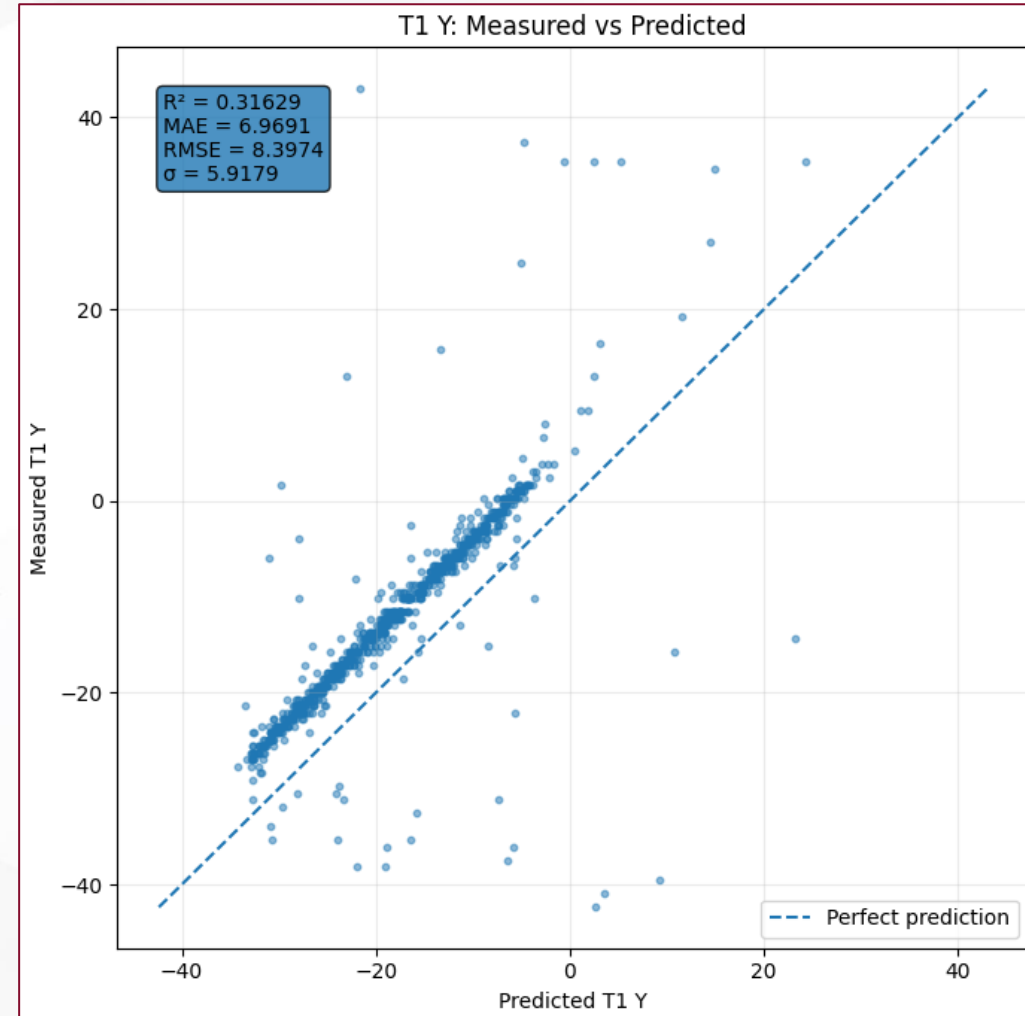
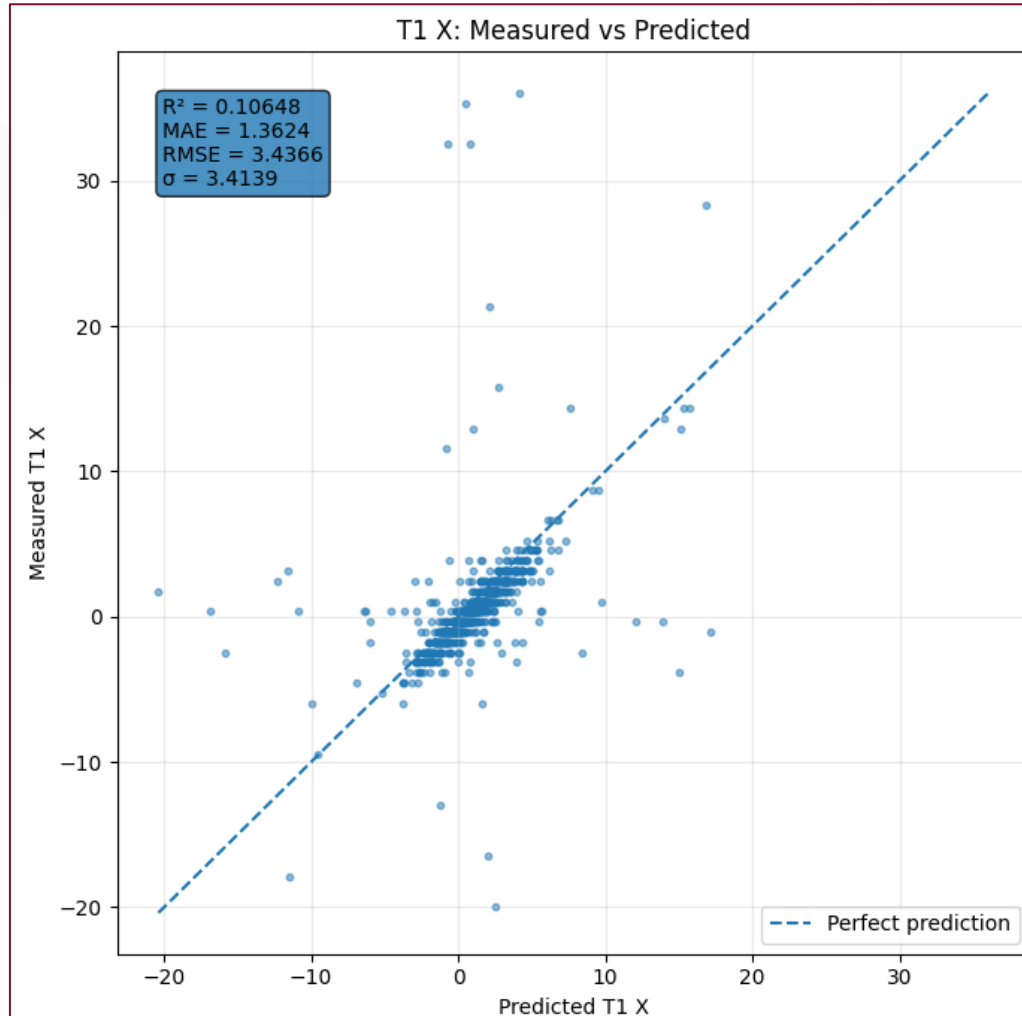
For each event:

- Fit trajectories for the hit in the xz and yz planes
- Use trajectories to calculate det5 expected position
- Analyze stats on the residuals

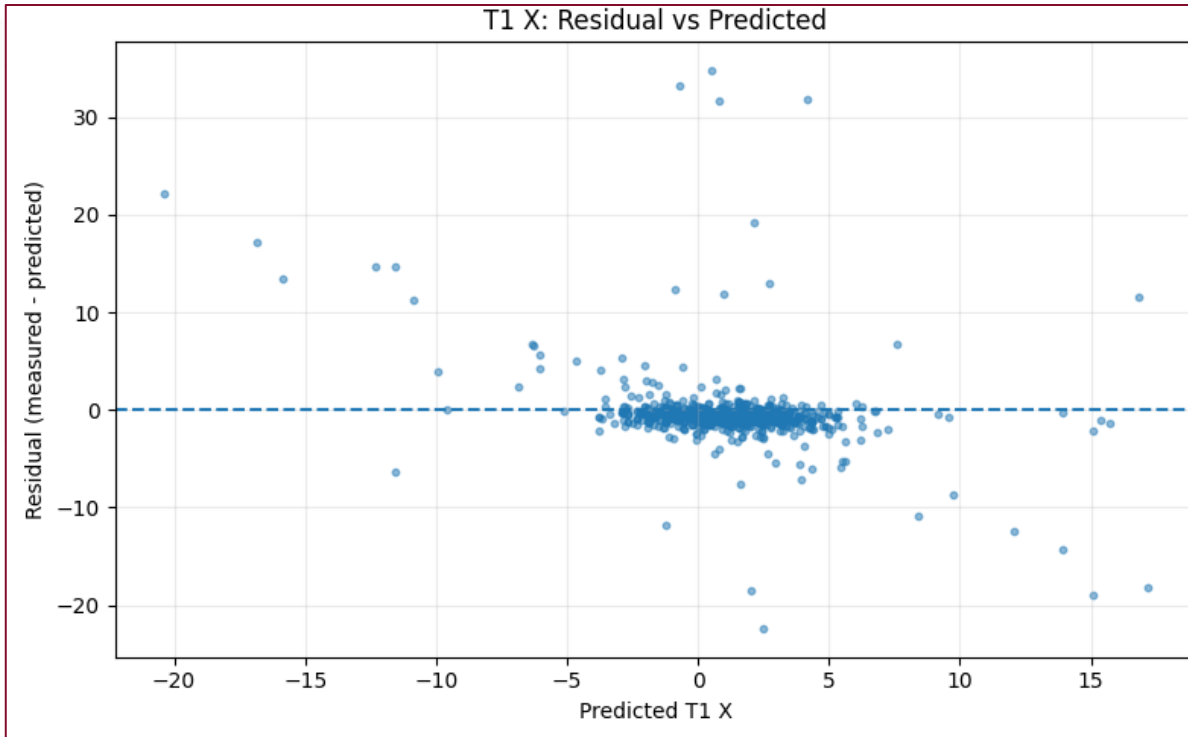
Cleaning and baseline:

- Fit trajectories for the hit in the xz and yz planes using 3/4 trackers
- Use trajectories to calculate expected position of left out tracker
- Analyze stats on the residuals
- Apply translation for average residual

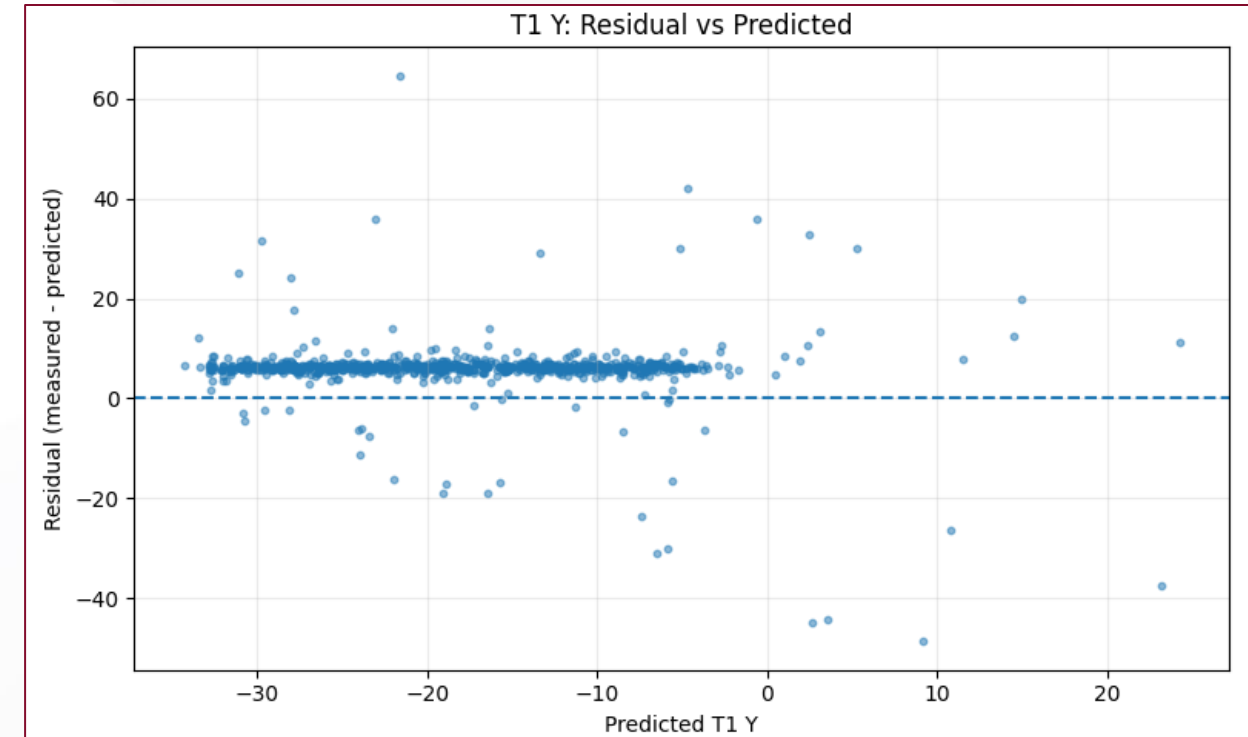
T1 Residuals



T1 Residuals

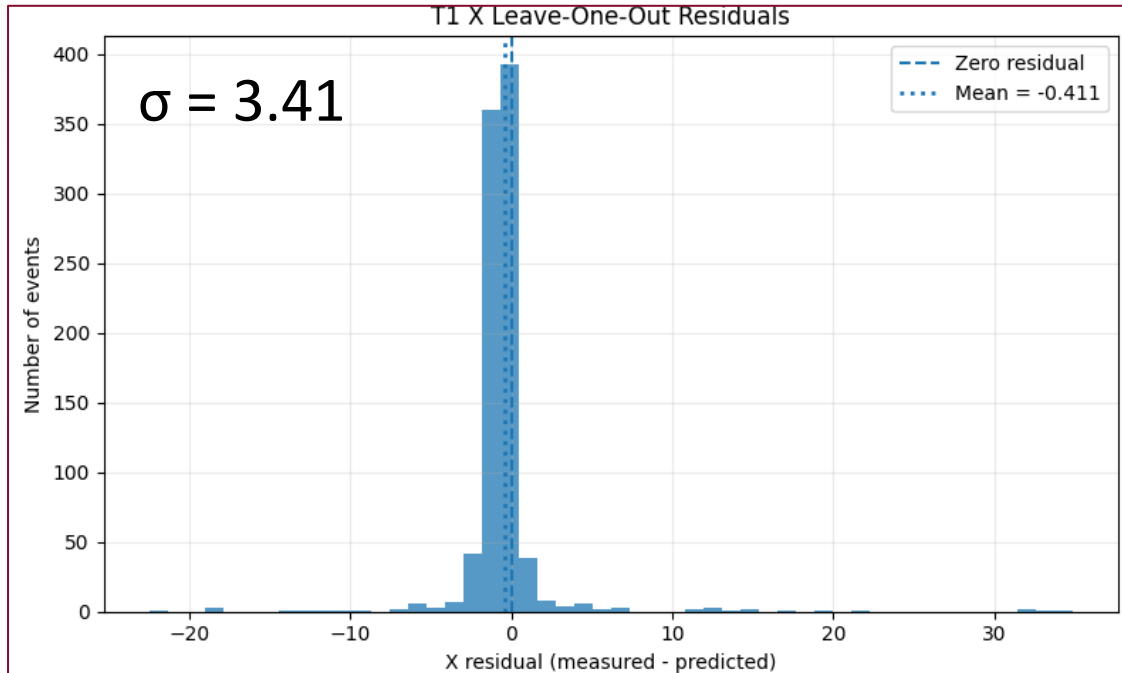


T1 X: mean residual = -0.410962, correction = 0.410962

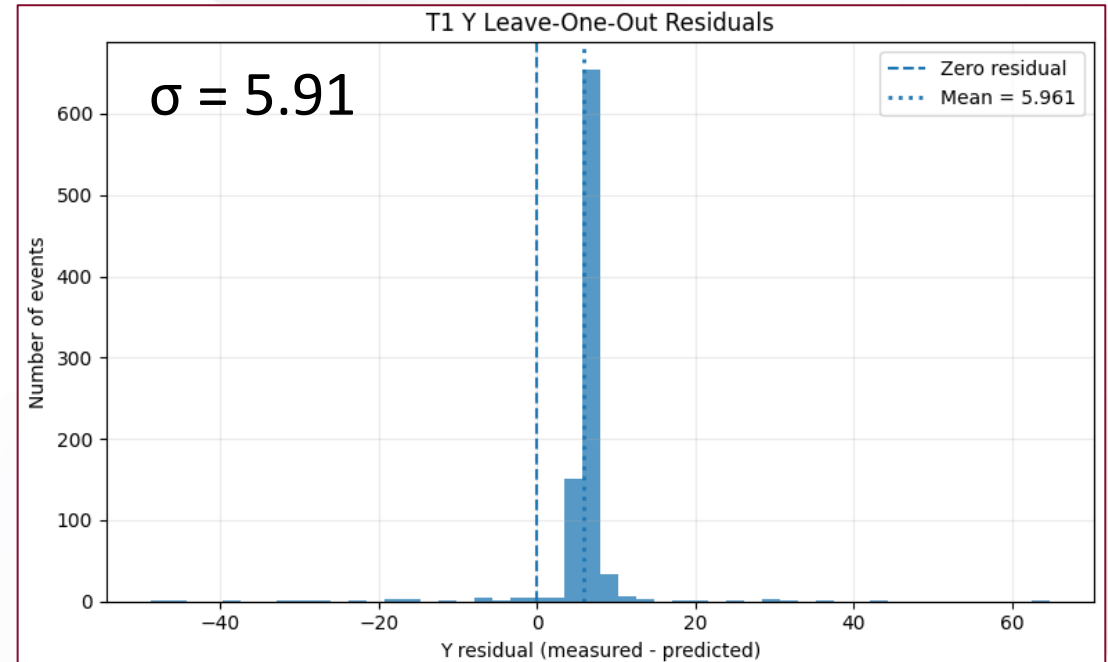


T1 Y: mean residual = 5.961020, correction = -5.961020

T1 Residuals



T1 X: mean residual = -0.410962, correction = 0.410962



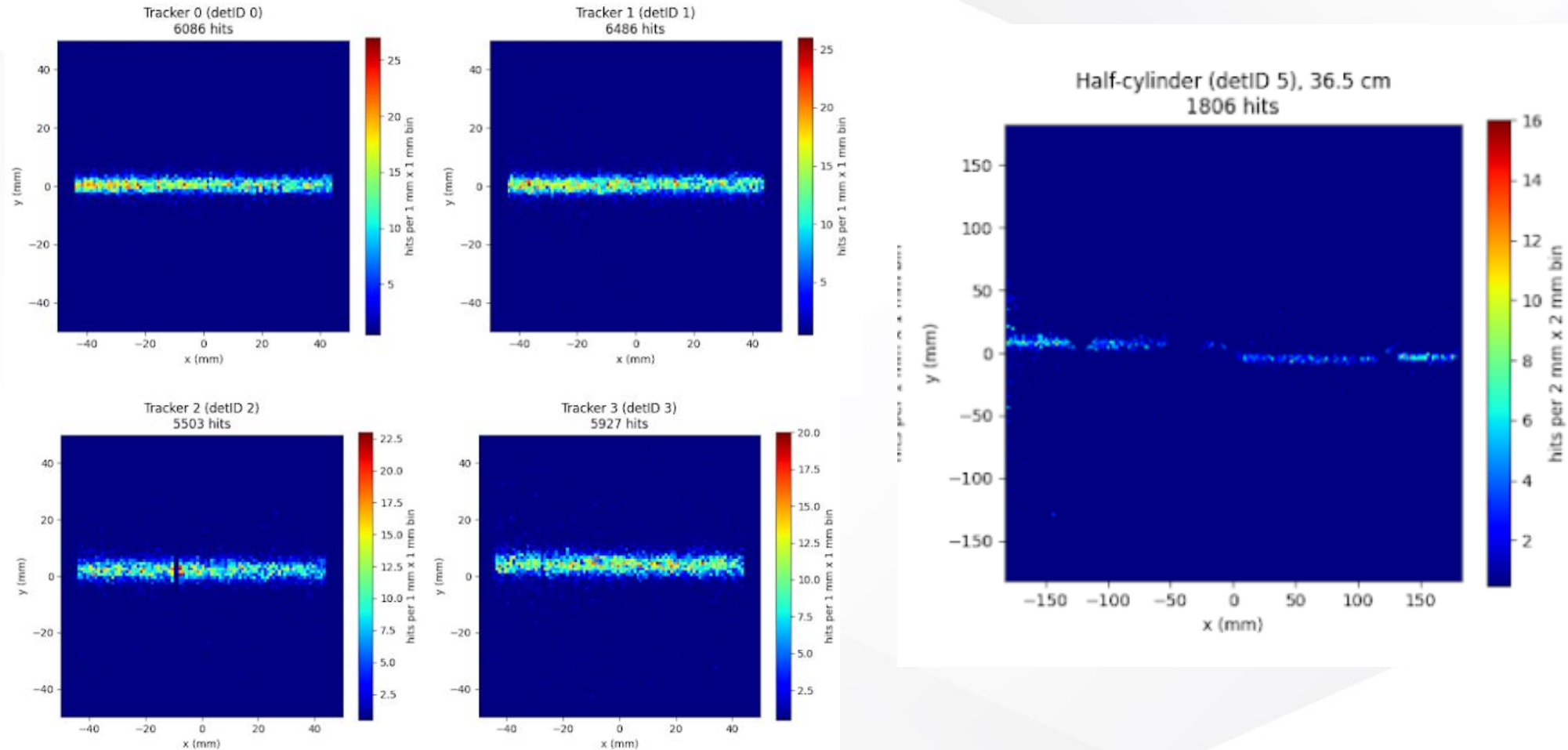
T1 Y: mean residual = 5.961020, correction = -5.961020

Efficiency Plots Demo Video

```
(uproot-env) summe@summerlenovo:~$ python3 tracker_residuals.py tracker_positions_adc_100.csv 0
```

<https://youtu.be/WT2Xoli7nTI>

Task- Compare Cylindrical Detector to Trackers



M5 Tasks

Perform and verify better data preprocessing:

- Dynamic adc threshold
- Clustering
- Iterative alignment algorithm

Automate pipeline for multiple root files

Efficiency plots and other analysis:

- Apply revised residual pipeline to trackers and cylindrical detector
- Plots histograms for cluster multiplicity

Dynamic Thresholding and Clustering

Previous hit selection process:

- Select candidate strips where $\max(\text{adc}) > \text{SET_THRESHOLD}$
- Static threshold set to 100 typically
- Calculate positions from raw strip values

Revised hit selection process to be applied:

- Calculate μ_{noise} and σ_{noise} for each strip from pedestal run
- Select candidate strips where for each strip $\max(\text{adc}) > \mu_{\text{noise}} + 5\sigma_{\text{noise}}$
- Cluster the candidate strips with multiplicity in range 2-10
- Calculate positions from clusters

Milestone 5 Task Matrix

Task	Alejandro	Summer
I. Apply clustering and iterative offset shift before reconstructing	50%	50%
II. Automate beam reconstruction pipeline for multiple root files	50%	50%
III. Do efficiency plots and other requested graphs for analysis	50%	50%

Thank you for listening

Questions?

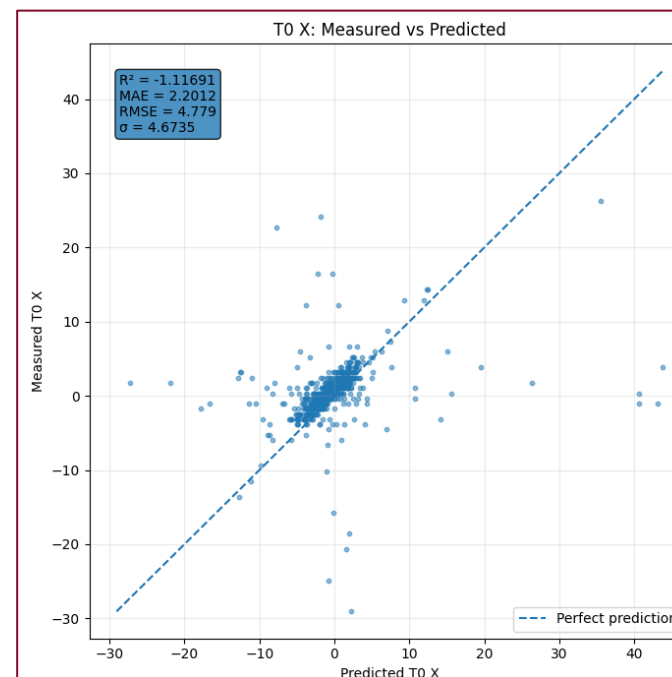
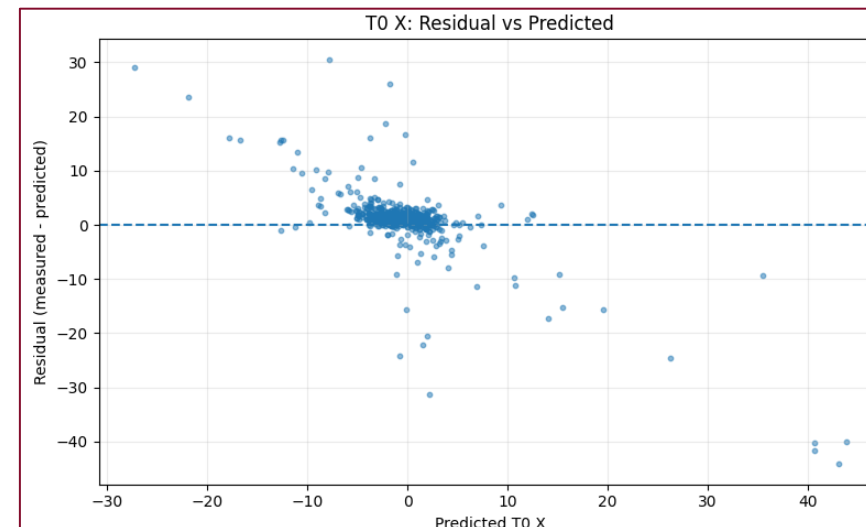
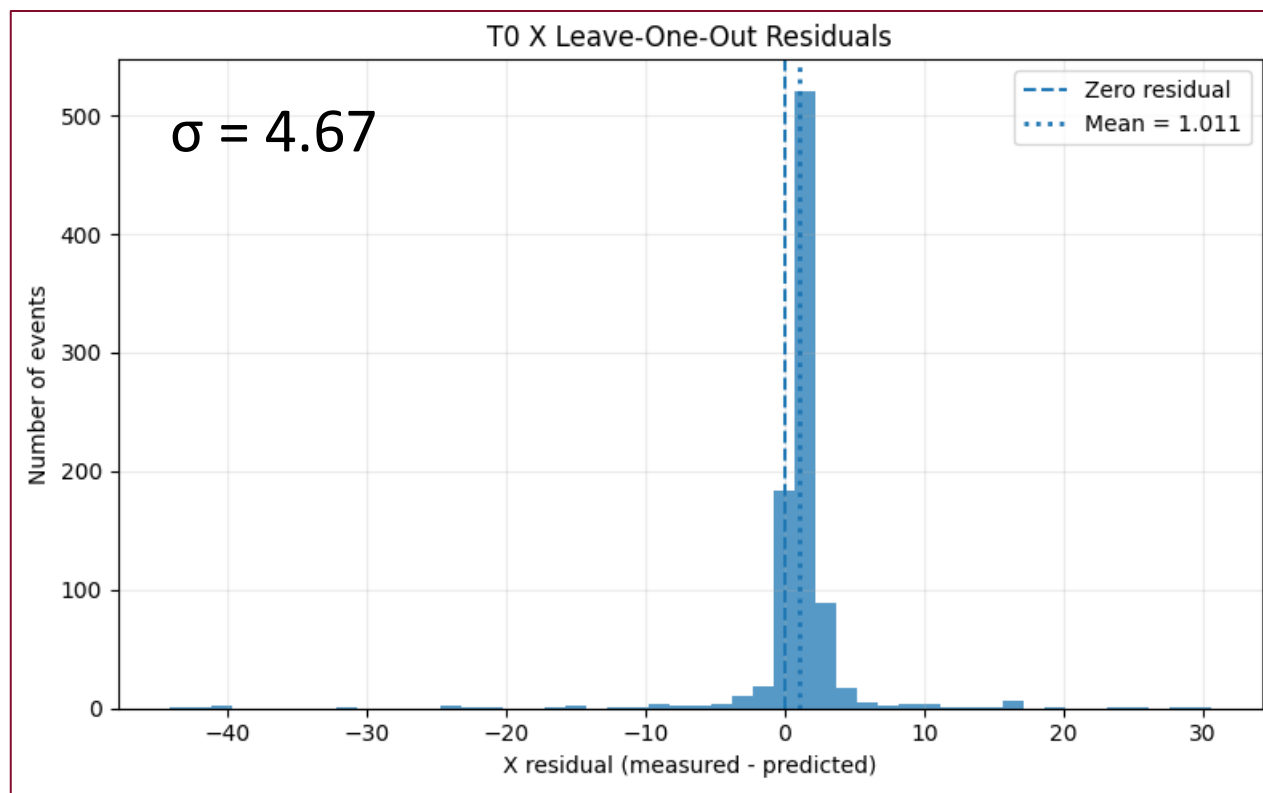
Definitions

- **Detector-** A detector is a device used in particle physics to measure the position of charged particles.
- **μ RWELL (Cylindrical micro-resistive well detector)-** μ RWELL is a type of micro-pattern gaseous detector that combines gas electron multiplication with a resistive protection layer and a segmented readout structure to measure charged particles with high precision [1]. Designing a detector that is cylindrical is the novelty here.
- **Tracker-** In the context of this project, a tracker refers to an external planar detector. Four trackers are used to create a reference system for evaluating the efficiency of the cylindrical detector.
- **Readout board-** A readout board is an electronic circuit board that collects signals from detector channels. The readout board hosts APV chips and interfaces between the detector strips and the data acquisition system. Each readout board can only capture a single plane of the data. So in planar detectors, typically there is one for x and y positions, respectively.
- **Channels/strips-** Channels or strips are the readout elements on the detector surface that collect charge from ionization events. Each strip is connected to a specific APV channel. Strip signals are used to reconstruct particle hit positions.
- **APV chip (Analog Pipeline Voltage chip)-** An APV chip samples analog signals from each channel on the readout board in time bins.
- **Time bin-** A time bin is a discrete time sample recorded by the APV chip. Typically, 12-16 time bins are used to reconstruct the pulse shape of a beam.
- **ADC (Analog to Digital Converter value)-** ADC represents a numerical value for the current recorded on a particular strip.

Definitions cont.

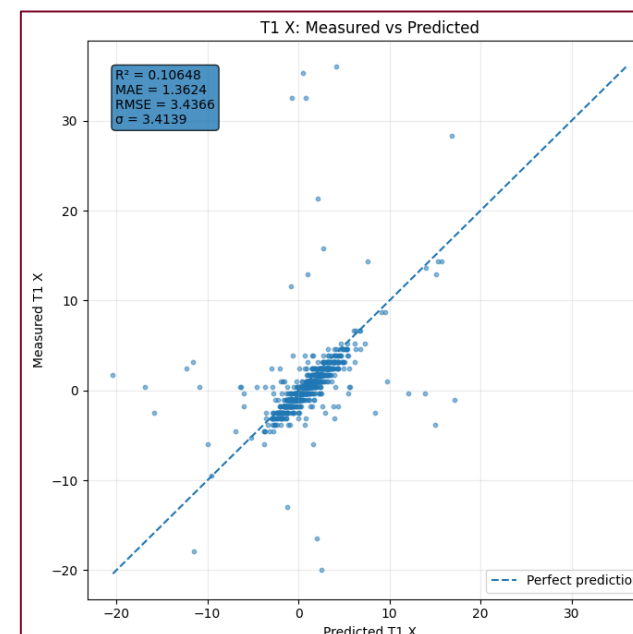
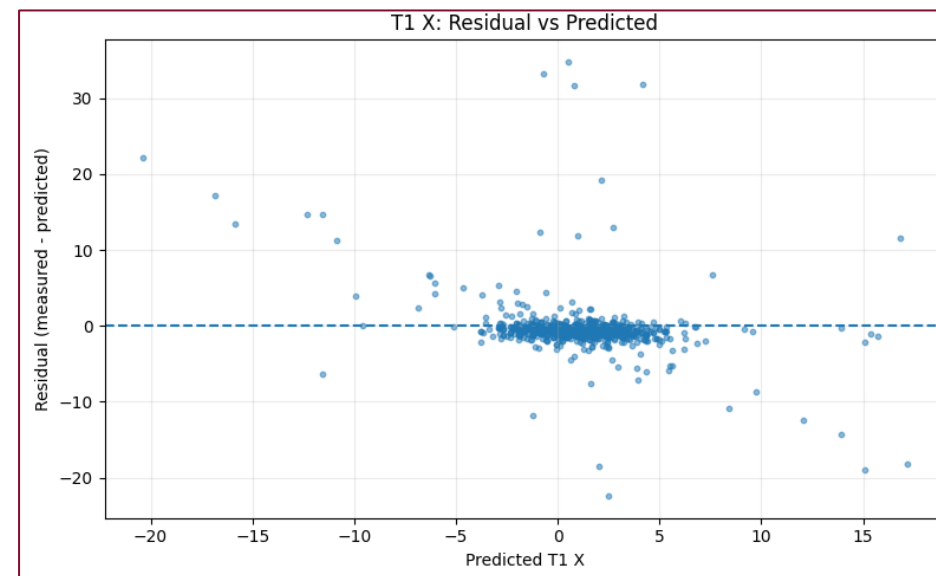
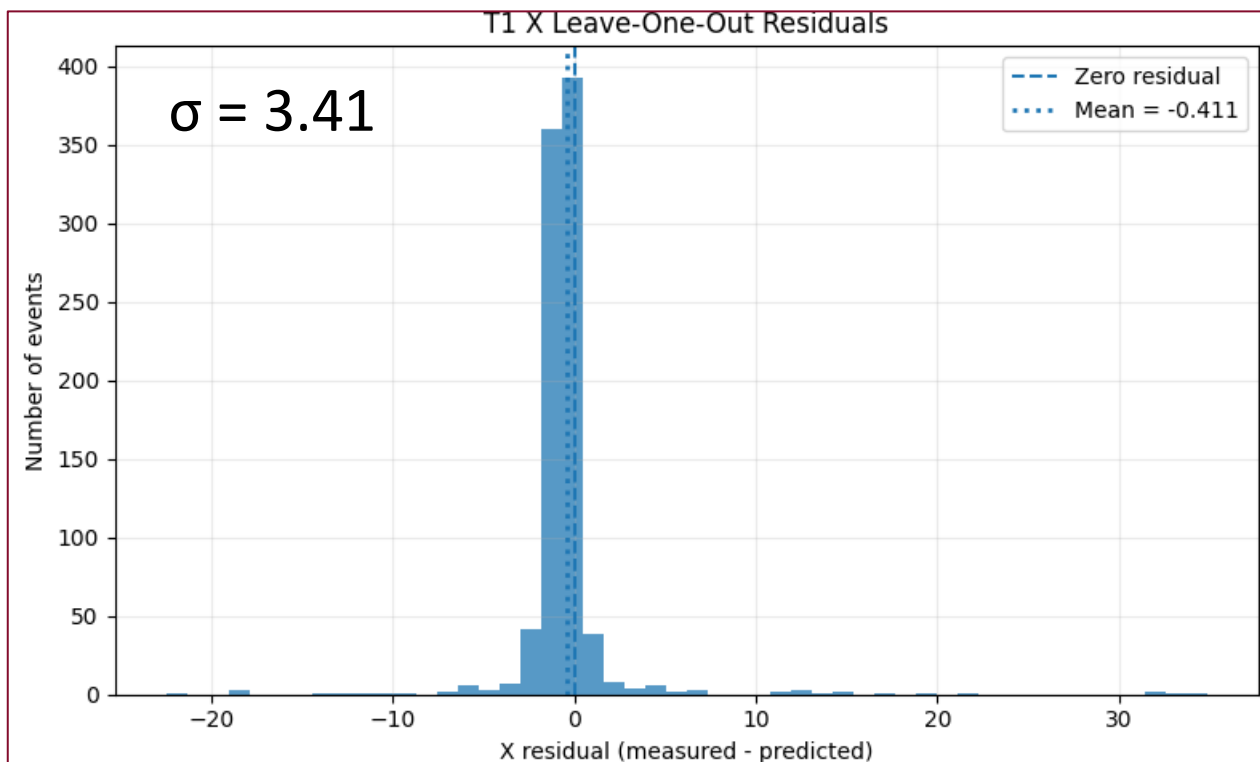
- **AMORE-SRS (Analysis and Monitoring Of Readout Electronics and Scalable Readout System)**- AMORE-SRS is a live data acquisition framework for monitoring and organizing the data from the APV chips into a root file for analysis.
- **Root file**- The root file stores event data such as strip information, ADC values, and metadata. The data is organized into branches that correspond to specific variables.
- **3D Pulse plot**- A 3D pulse plot is a visualization of a single event where the x axis is time bin, the y axis is the strip number, and the z axis is the ADC value.
- **Resolution plot**- A resolution plot is a reconstruction of the 2d hit positions represented as a heat map.
- **Efficiency plot**- Efficiency plots compare the detectors efficiency against other performance characteristics such as amplification or drift field. Efficiency itself is defined as the ratio of the number of events with strip clusters seen in all five detectors divided by the number of events with strip clusters seen in just the four trackers [2].

T0 x Residuals



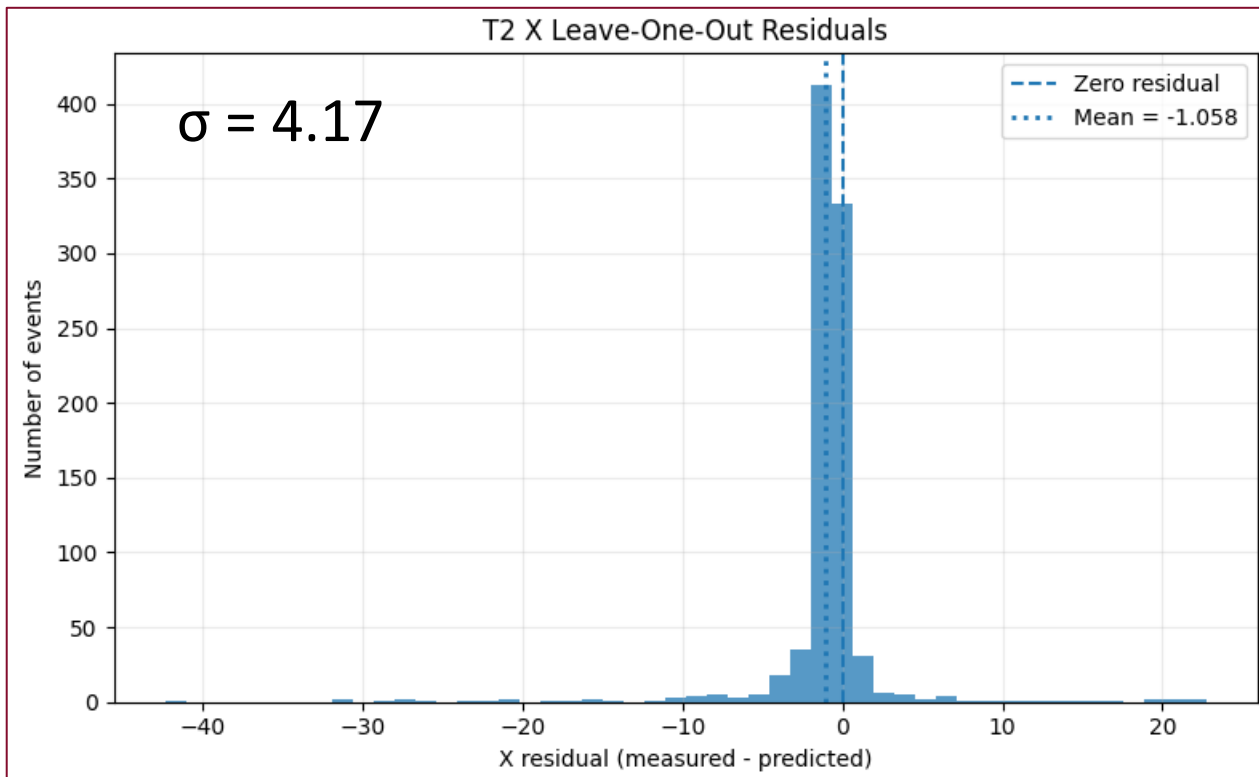
T0 X: mean residual = 1.010579, correction = -1.010579

T1 x Residuals

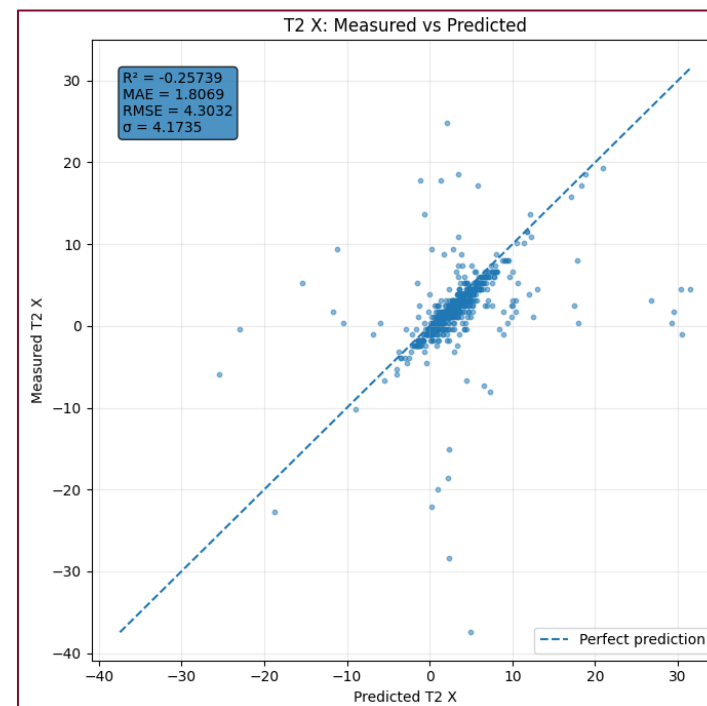
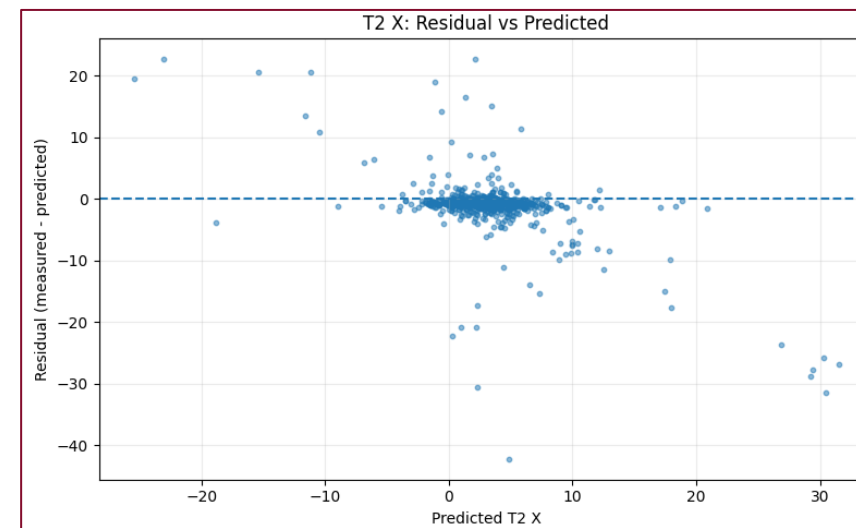


T1 X: mean residual = -0.410962, correction = 0.410962

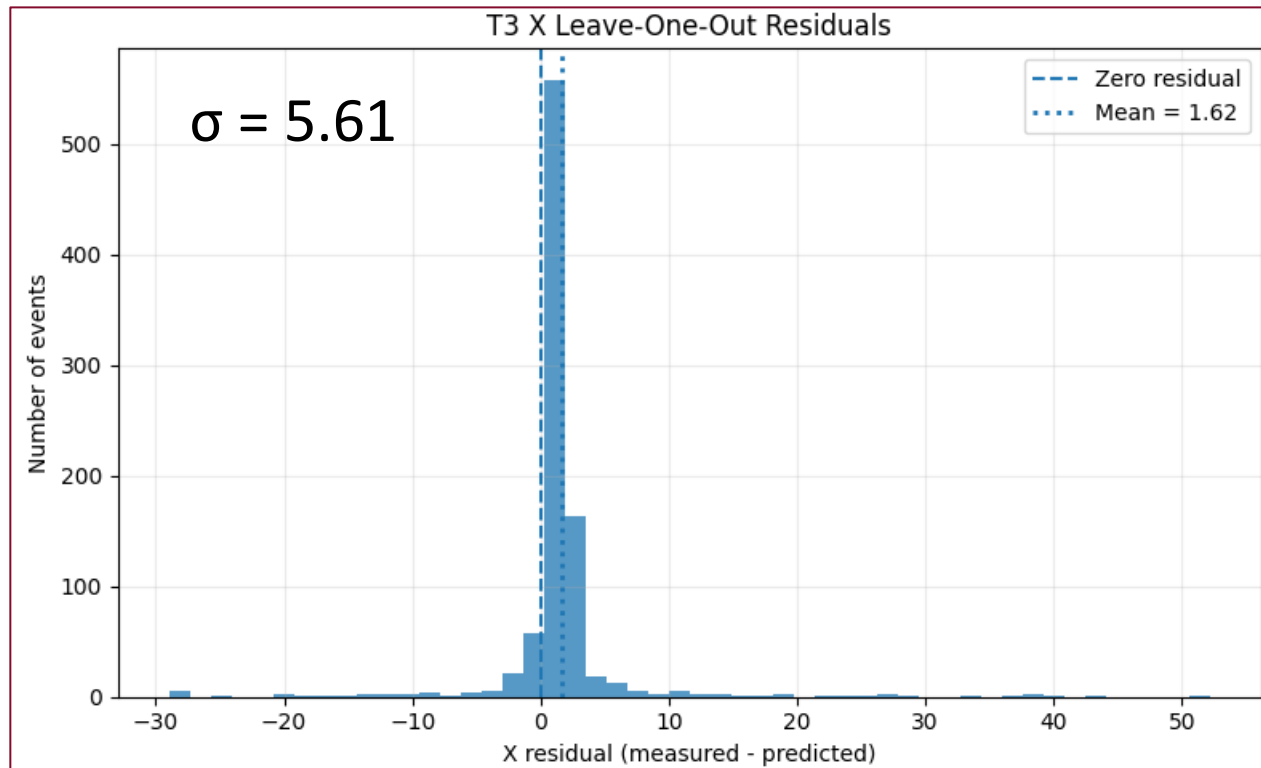
T2 x Residuals



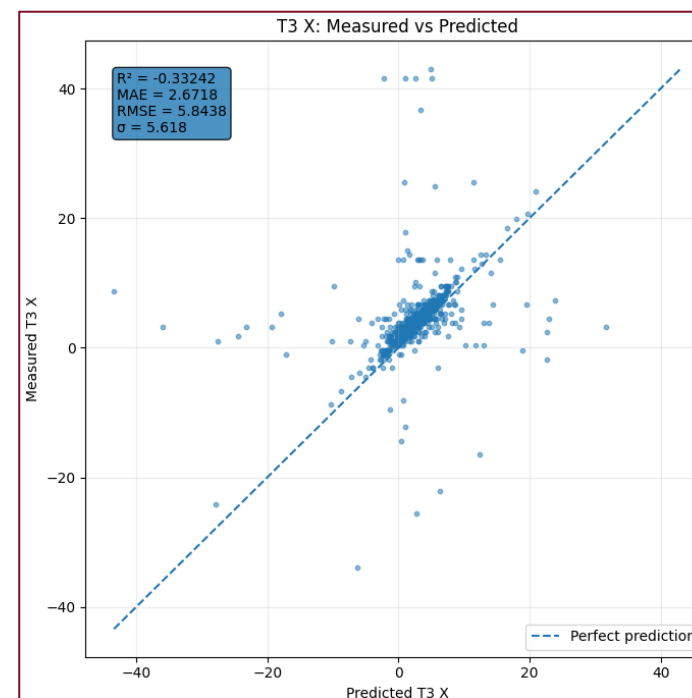
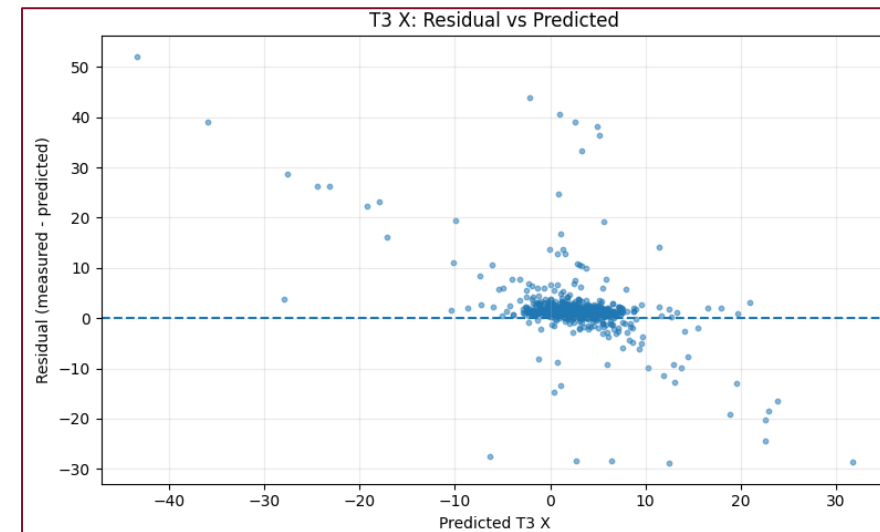
T2 X: mean residual = -1.057699, correction = 1.057699



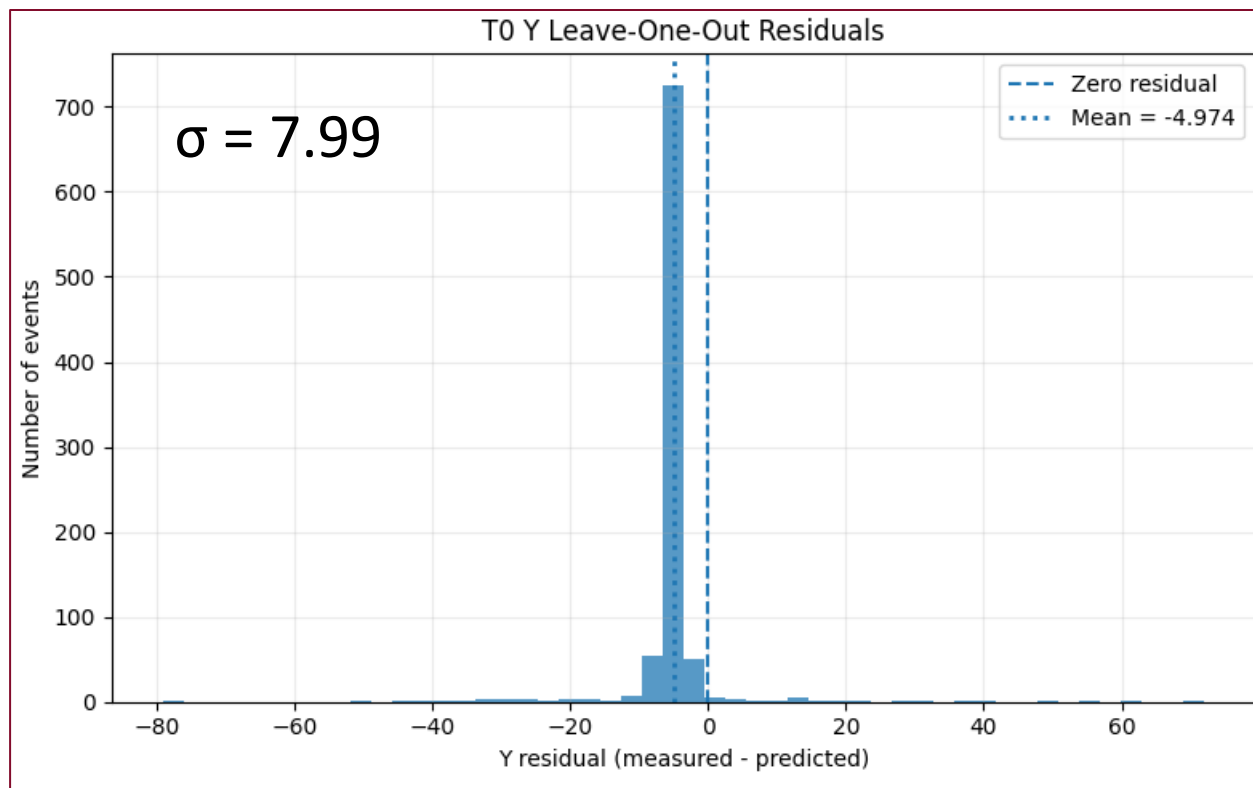
T3 x Residuals



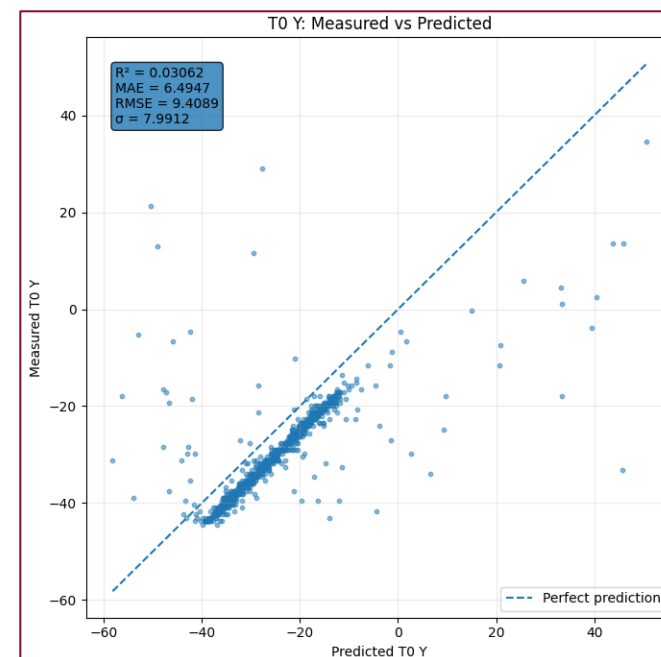
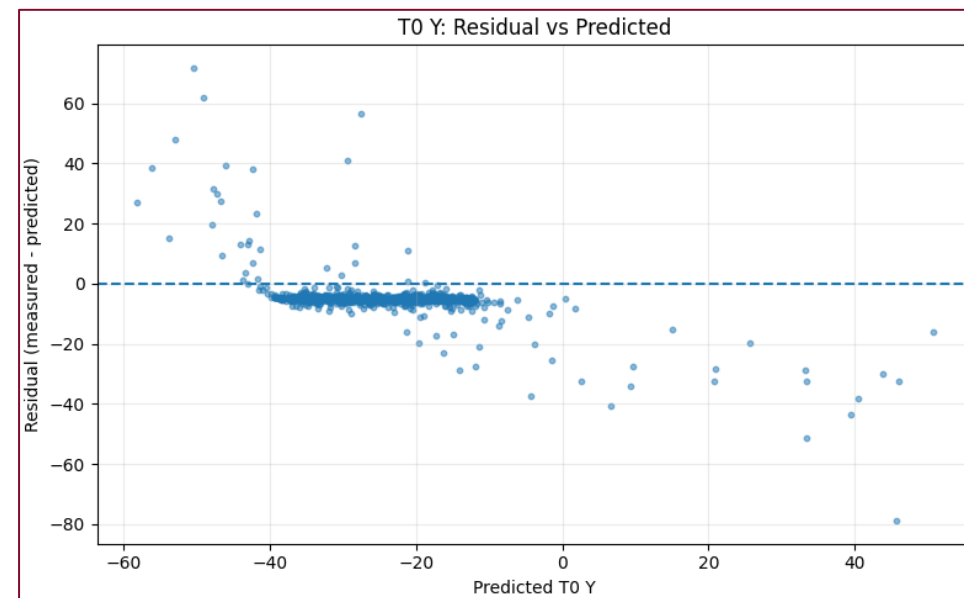
T3 X: mean residual = 1.619634, correction = -1.619634



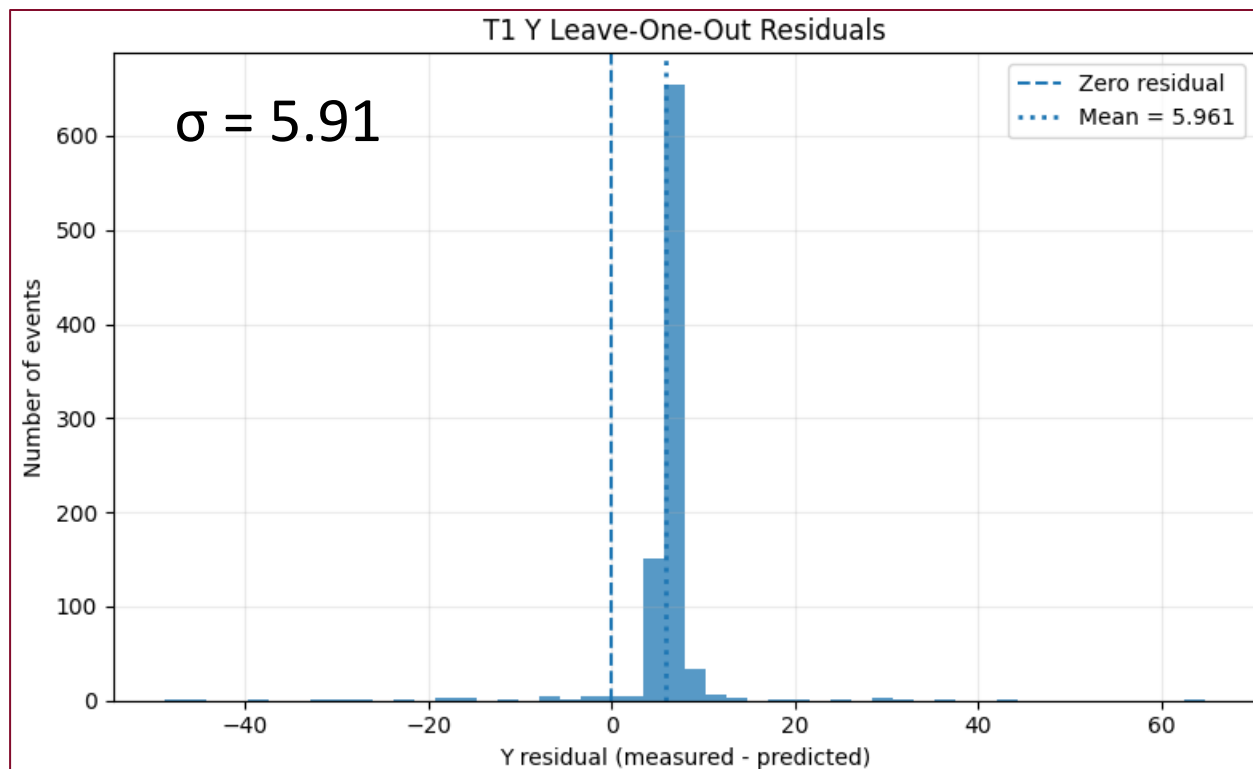
T0 y Residuals



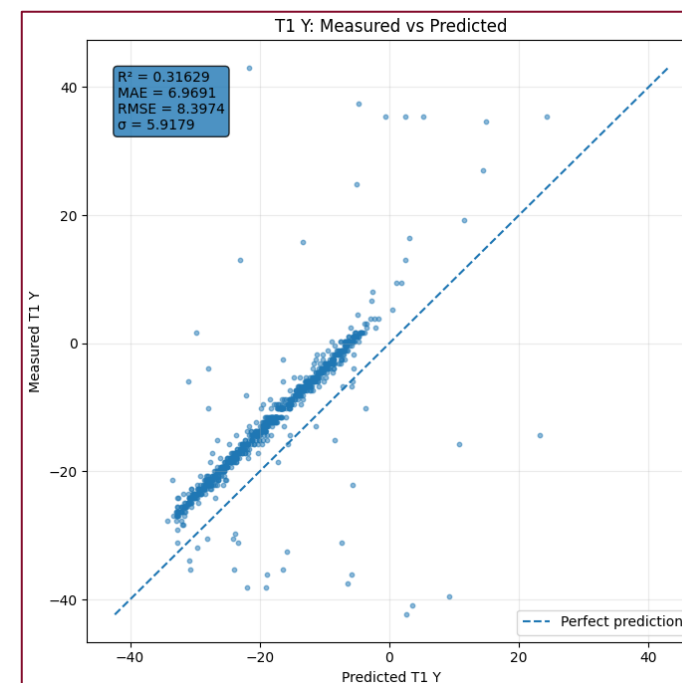
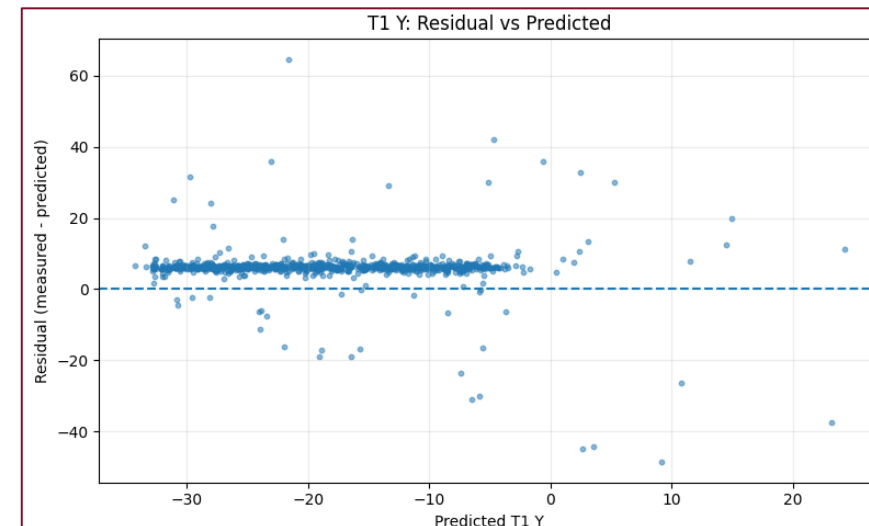
T0 Y: mean residual = -4.973888, correction = 4.973888



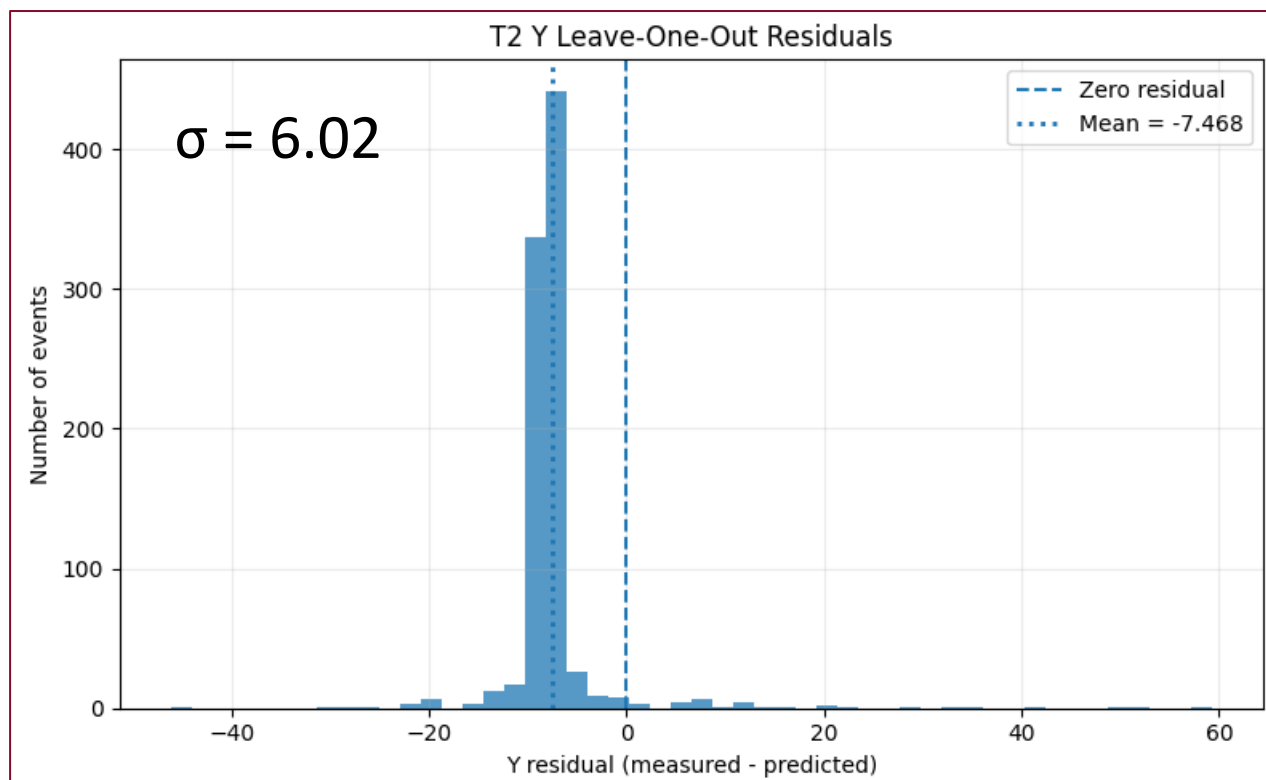
T1 y Residuals



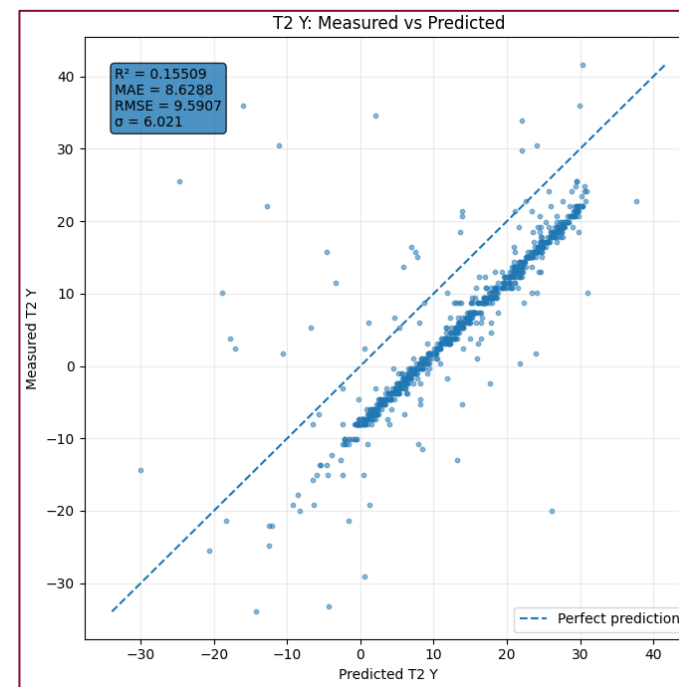
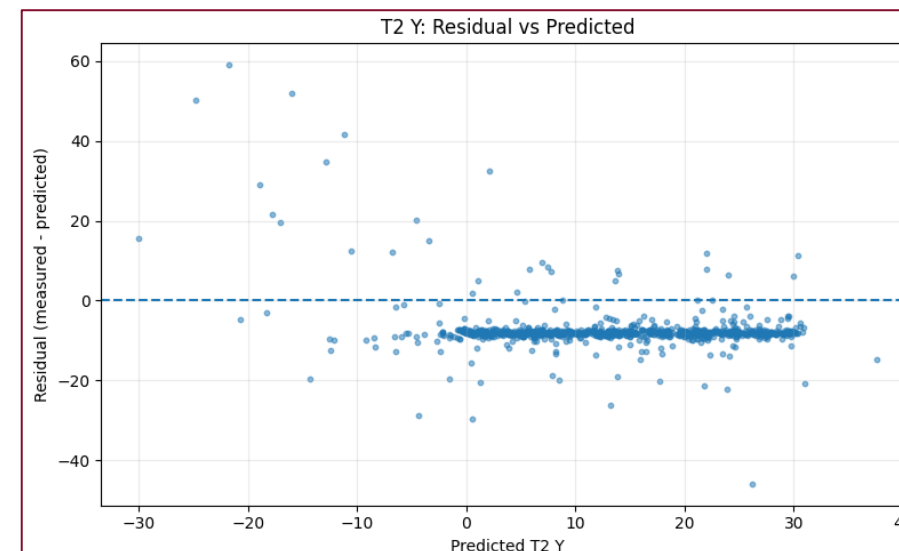
T1 Y: mean residual = 5.961020, correction = -5.961020



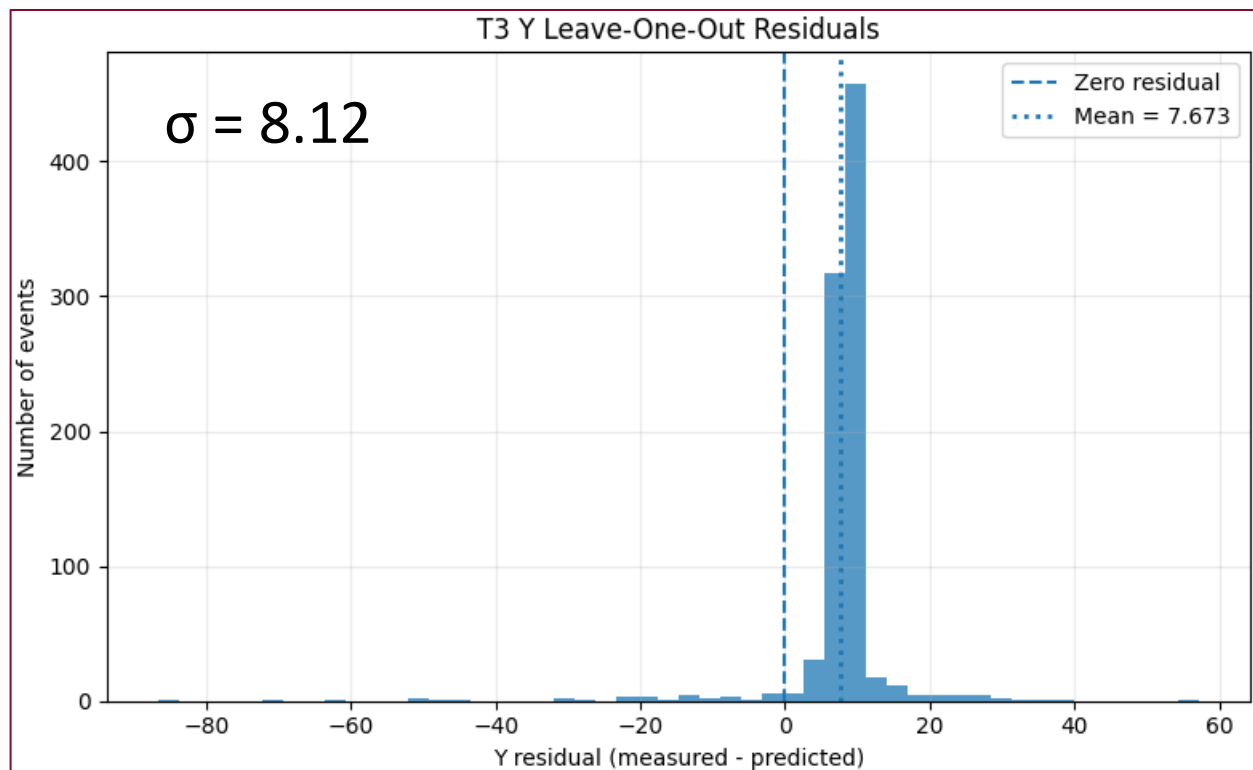
T2 y Residuals



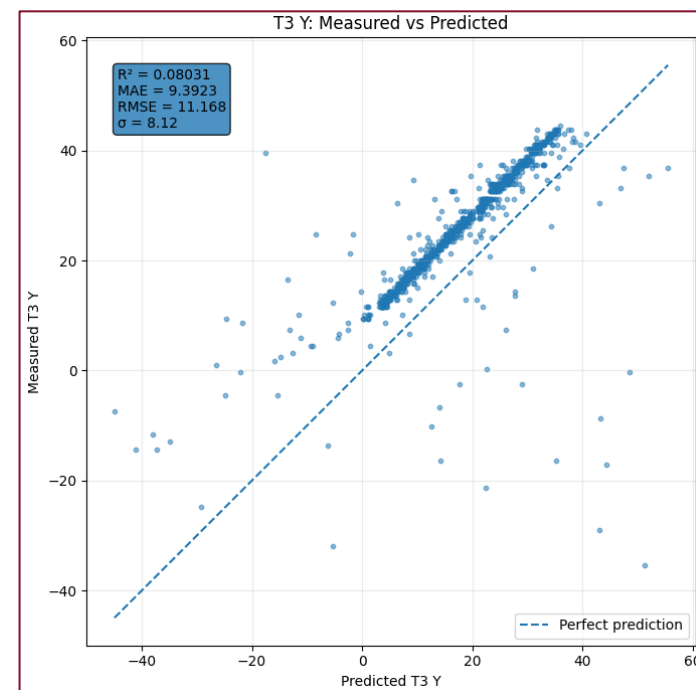
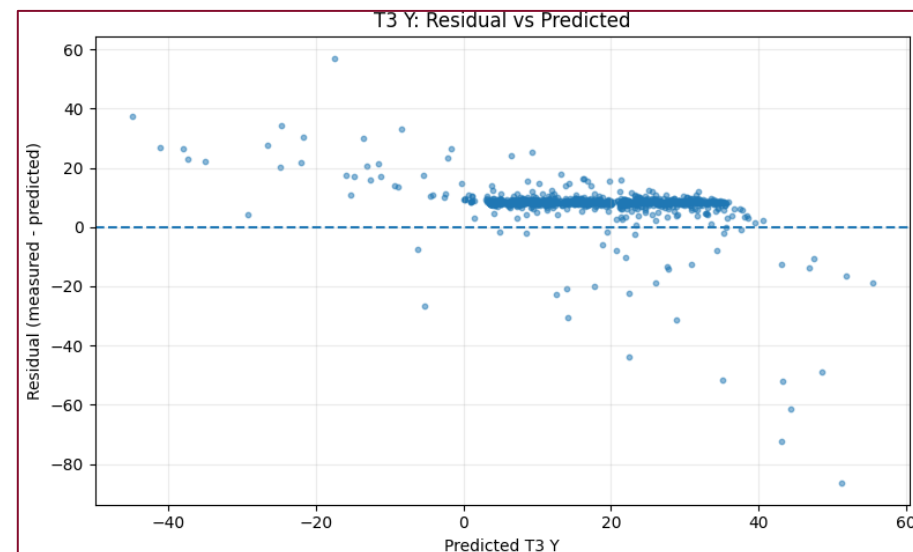
T2 Y: mean residual = -7.467947, correction = 7.467947



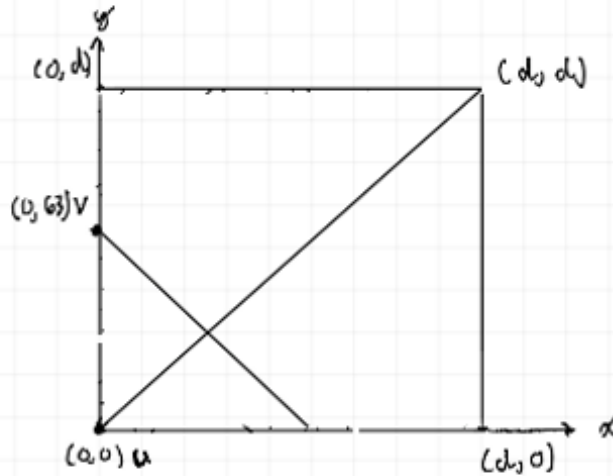
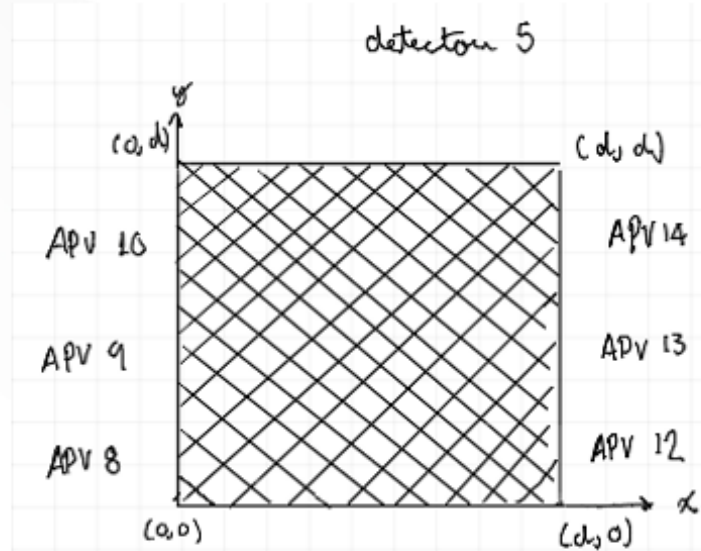
T3 y Residuals



T3 Y: mean residual = 7.672630, correction = -7.672630



Approach- Resolution Plot



$$p_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad d_1 = u = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$p_2 = \begin{bmatrix} 0 \\ d_3 \end{bmatrix} \quad d_2 = v = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$l_1 = p_1 + t d_1 \quad l_2 = p_2 + \lambda d_2$$

$$p_1 + t d_1 = p_2 + \lambda d_2$$

$$t d_1 - \lambda d_2 = p_2 - p_1$$

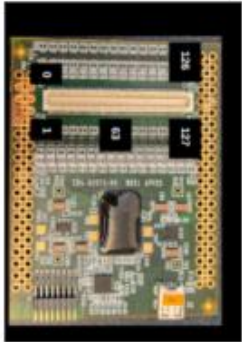
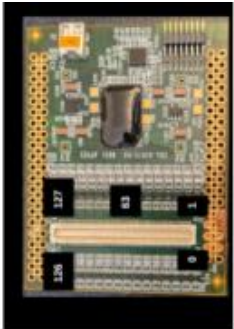
$$\begin{bmatrix} d_{1x} & -d_{2x} \\ d_{1y} & -d_{2y} \end{bmatrix} \begin{bmatrix} t \\ \lambda \end{bmatrix} = \begin{bmatrix} p_{2x} - p_{1x} \\ p_{2y} - p_{1y} \end{bmatrix}$$

Channel Orientations

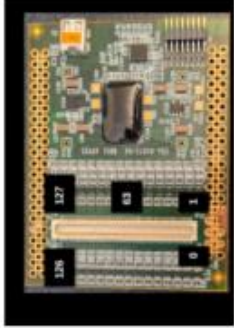


Physical Local Strip Order

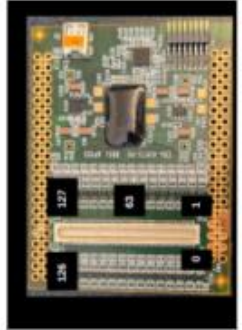
1	127
127	1
126	0
0	126



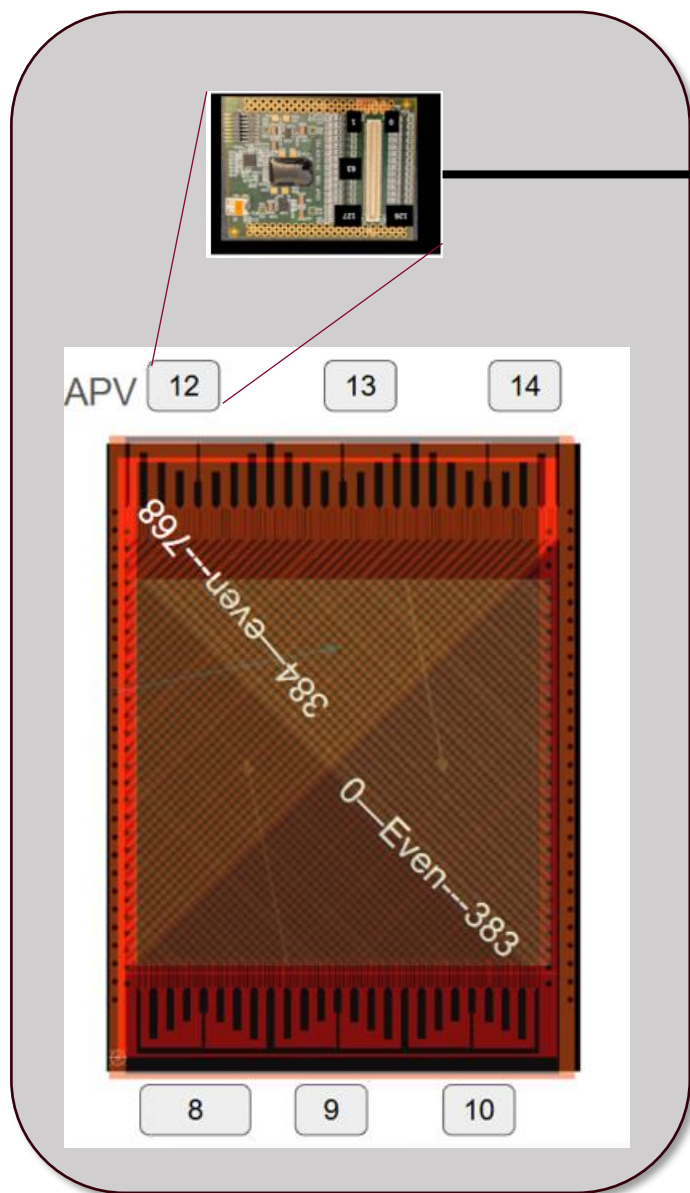
<u>126</u>	<u>0</u>
<u>0</u>	<u>126</u>
<u>1</u>	<u>127</u>
<u>127</u>	<u>1</u>



1	127
127	1
126	0
0	126



Hardware



Software

